AWS Networking Workshop

- Foundational Topics
  - Prerequisites
  - VPC Fundamentals
  - Multiple VPCs
  - Security Controls
  - Connecting to On-Premises
    - Establish VPN Connection
    - Explore Hybrid Environment
    - Hybrid DNS
    - Clean Up
  - Connecting to SDWAN
  - Network Monitoring
  - Network analytics using Amazon Quick Suite
- Advanced Topics
- Contributors & Revision History

Workshop catalog in AWS Builder Center

Content preferences

Language

English

Connecting to On-Premises

Please note, this module is optional and is not required to be completed to proceed with this workshop. You may skip to Network Monitoring

Elastic IP quota
This module's stack creates two Elastic IP addresses. If the **Connecting to SDWAN** stack is still deployed in this account, first complete only the SDWAN-specific steps of the SDWAN cleanup — through deleting the `ThirdPartyAppliance-TGWConnect` stack, stopping before "Continue deleting resources", which removes shared stacks this module still needs. Running both modules at once exceeds the default quota of five Elastic IPs per Region and this stack will fail to deploy.

Amazon Virtual Private Cloud (Amazon VPC) provides multiple options to integrate existing data center networks with your AWS VPCs, including AWS Managed Site-to-site (IPsec) VPNs and Direct Connect connections. AWS VPN connections provide up to **5 Gbps** of throughput per tunnel. When used in conjunction with the Transit Gateway, AWS VPN supports Equal-Cost Multipathing (ECMP) to allow you to scale VPN throughput.

In this lab, we will be standing up a simulated data center environment and connecting it with the existing Transit Gateway that was set up in the "Multiple VPCs" lab. Recall that at the end of that lab, we had provisioned 3 AWS VPCs and EC2 instances in each VPC. The VPCs were interconnected using Transit Gateway.

To build out our simulated data center environment and connect it to our AWS environment, we will:

1. Deploy a VPC containing a simulated datacenter environment, with DNS server and a simple web application.
2. Establish VPN connectivity between the simulated datacenter and the AWS environment.
3. Set up DNS resolution between the AWS environment and simulated datacenter.
4. Test connectivity from the AWS environment to the simulated datacenter.

If you are running this lab at an **AWS-led event**, your facilitator has already set the region for you. If you are running it **self-service**, simply use the region you are working in. In either case, the region shown in the screenshots may not match your environment — this will not cause any problems.

If you are running this lab in **your own AWS Account**, it is recommended for all lab resources to be created in us-east-1 region so that the screenshots match your environment. This is not mandatory.

Prerequisites

▶ **If you have not completed the Multiple VPCs sections...**

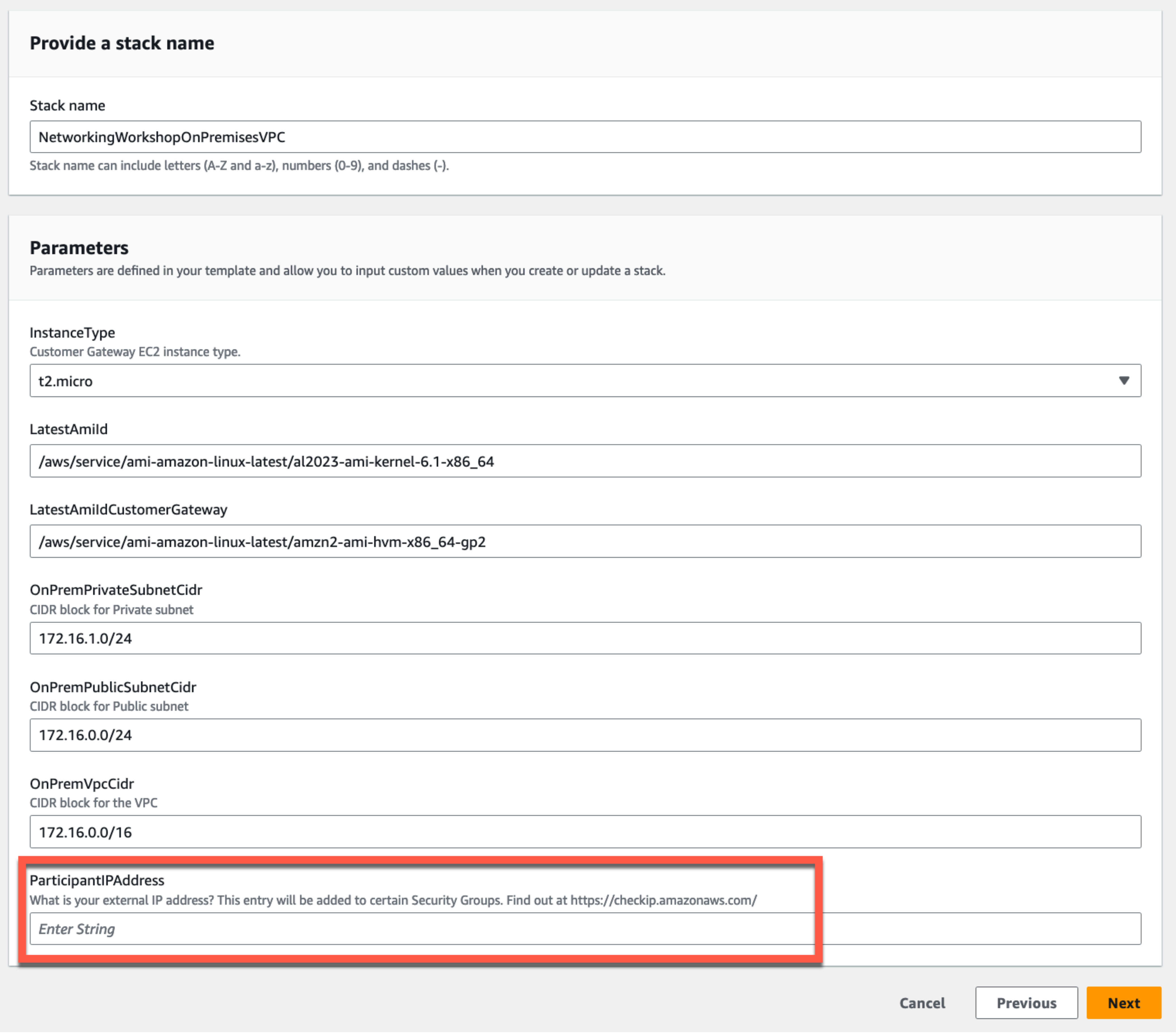
Deploy VPC to simulate 'On-Premises'

Now we are ready to deploy a provided CloudFormation template to create the simulated on-premises environment. This stack will create a VPC that will act as our on-premises environment with a public and private subnet as well as the three EC2 instances for a Customer Gateway Server, DNS Server, and Application Server.

1. Load the following CloudFormation template:

   CloudFormation Template

2. Enter the **Stack name** `NetworkingWorkshopOnPremisesVPC`
3. Update **ParticipantIPAddress** and then accept the remaining default values in **Configure stack options**
4. Click on **Next**
5. Click on **Create stack**



Provide a stack name

Stack name: NetworkingWorkshopOnPremisesVPC

Stack name can include letters (A-Z and a-z), numbers (0-9), and dashes (-).

Parameters

Parameters are defined in your template and allow you to input custom values when you create or update a stack.

InstanceType
Customer Gateway EC2 instance type.
t2.micro

LatestAmiId
/aws/service/ami-amazon-linux-latest/al2023-ami-kernel-6.1-x86_64

LatestAmiIdCustomerGateway
/aws/service/ami-amazon-linux-latest/amzn2-ami-hvm-x86_64-gp2

OnPremPrivateSubnetCidr
CIDR block for Private subnet
172.16.1.0/24

OnPremPublicSubnetCidr
CIDR block for Public subnet
172.16.0.0/24

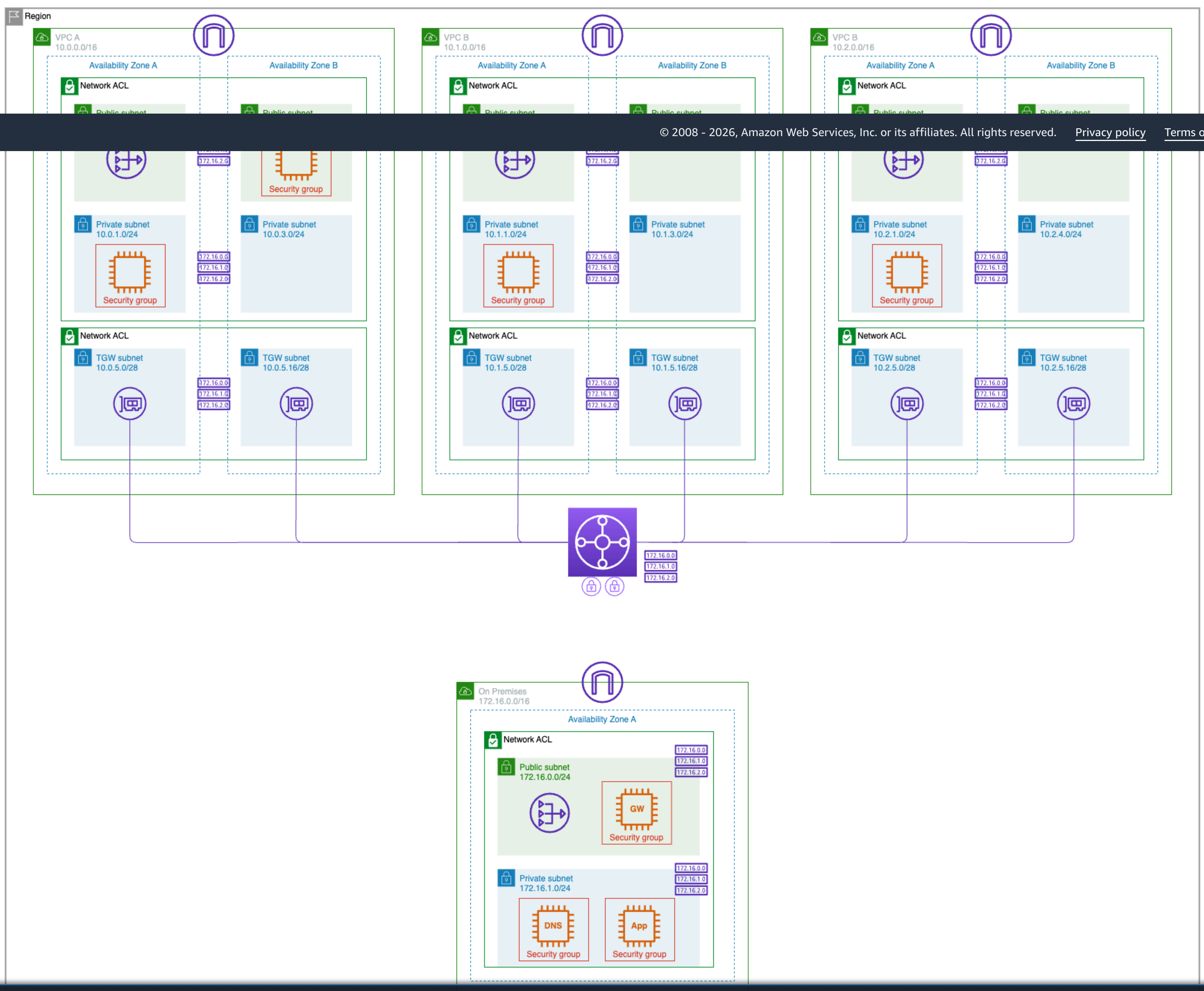
OnPremVpcCidr
CIDR block for the VPC
172.16.0.0/16

ParticipantIPAddress
What is your external IP address? This entry will be added to certain Security Groups. Find out at https://checkip.amazonaws.com/
Enter String

Wait for the stack to be created. This can take up to 10 minutes.

Important
If the deployment of the CloudFormation template fails due to the **Elastic IP address** limit being reached and you are using your own AWS account, please refer to the **Elastic IP Quota** section in Prerequisites - Your own AWS Account.

Now we are ready to create the VPN connection between AWS and the on-premise Customer Gateway



Select your cookie preferences

We use essential cookies and similar tools that are necessary to provide our site and services. We use performance cookies to collect anonymous statistics, so we can understand how customers use our site and make improvements. Essential cookies cannot be deactivated, but you can choose "Customize" or "Decline" to decline performance cookies.

If you agree, AWS and approved third parties will also use cookies to provide useful site features, remember your preferences, and display relevant content, including relevant advertising. To accept or decline all non-essential cookies, choose "Accept" or "Decline." To make more detailed choices, choose "Customize."

Accept
Decline
Customize

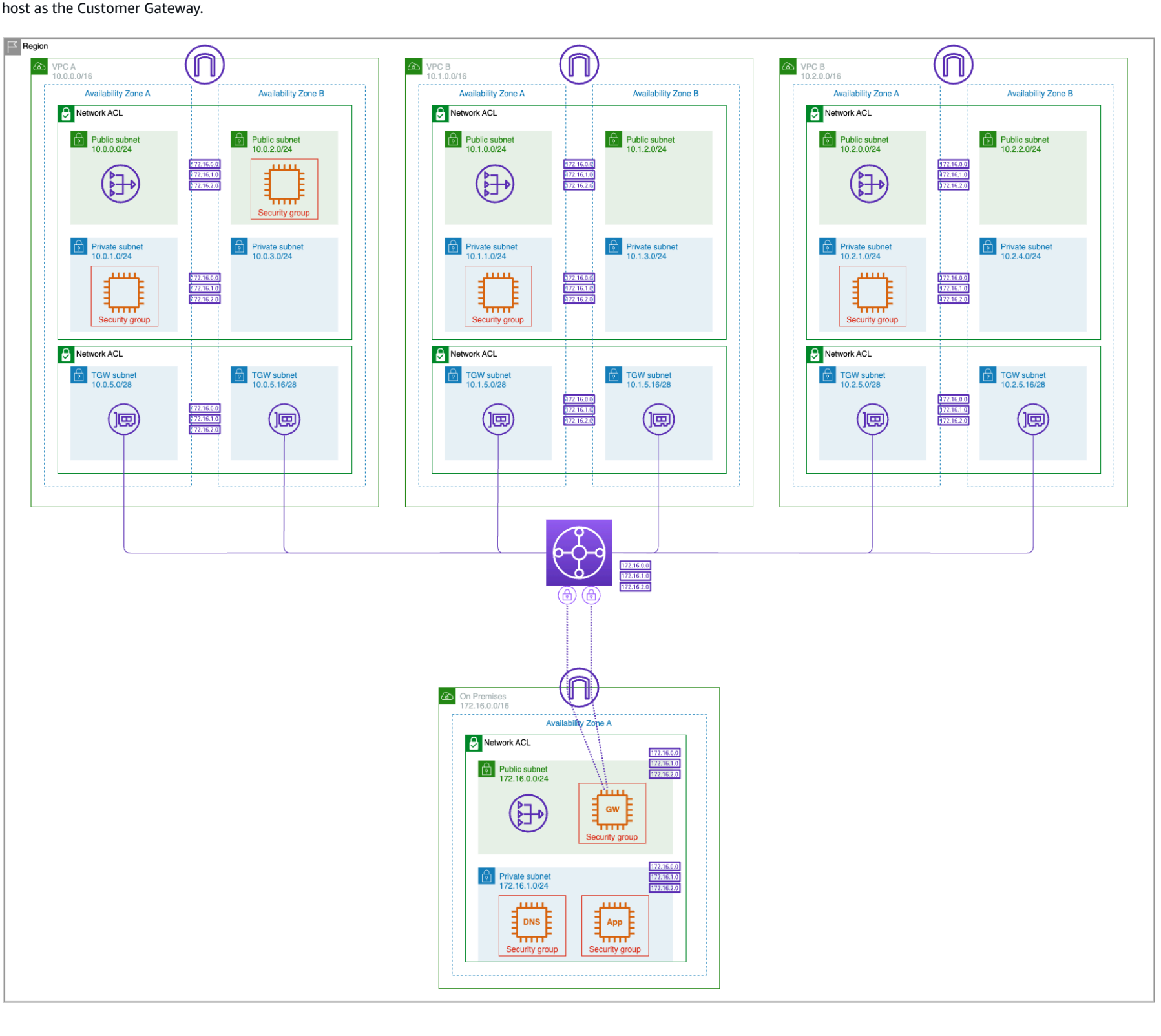
Select your cookie preferences

We use essential cookies and similar tools that are necessary to provide our site and services. We use performance cookies to collect anonymous statistics, so we can understand how customers use our site and make improvements. Essential cookies cannot be deactivated, but you can choose “Customize” or “Decline” to decline performance cookies.

If you agree, AWS and approved third parties will also use cookies to provide useful site features, remember your preferences, and display relevant content, including relevant advertising. To accept or decline all non-essential cookies, choose “Accept” or “Decline.” To make more detailed choices, choose “Customize.”

Establish VPN Connection

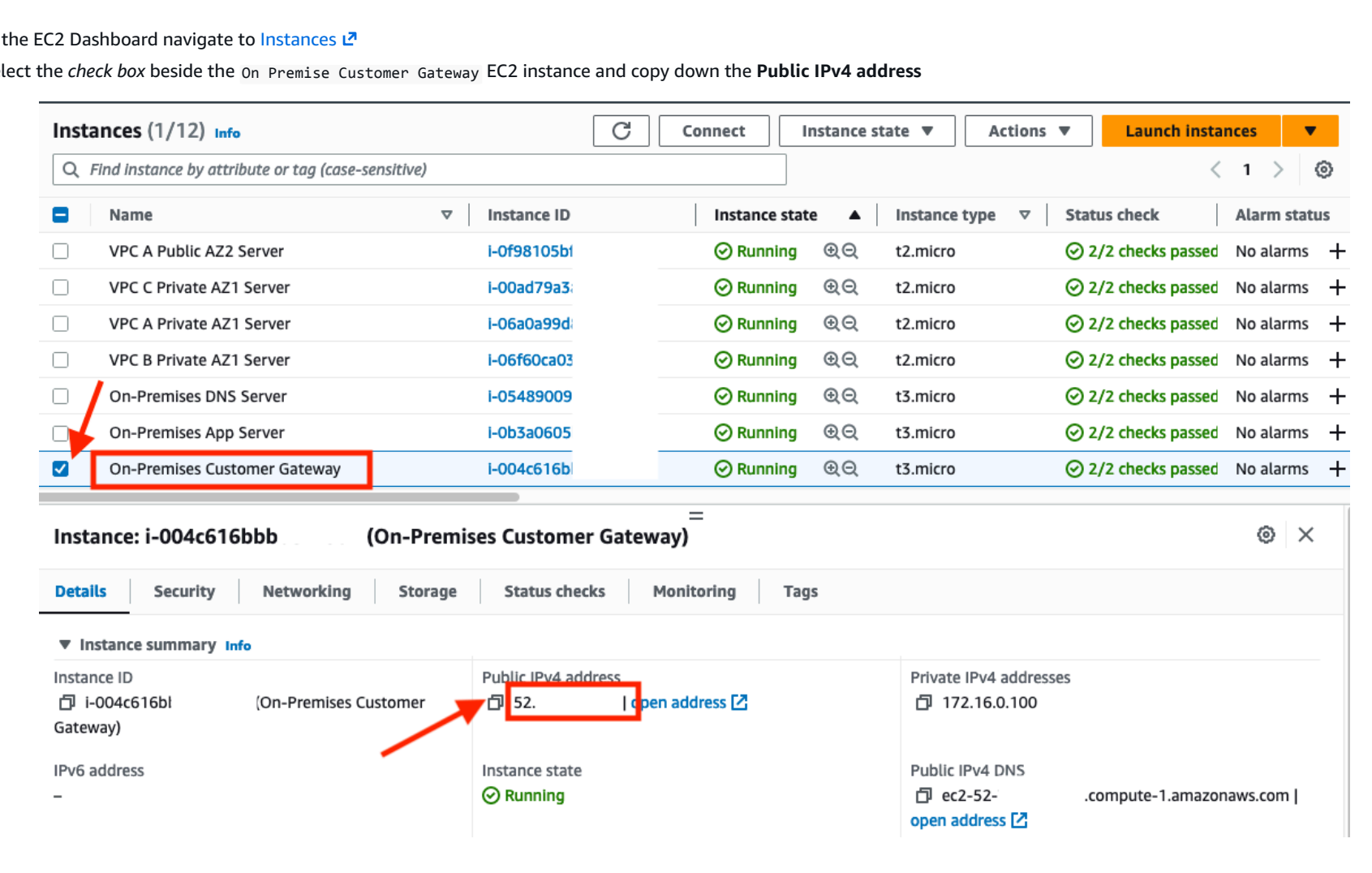
In the earlier Multiple VPCs lab we created a Transit Gateway to interconnect our the VPCs. To integrate the simulated datacenter environment, we will establish a VPN connection between the Transit Gateway and a customer gateway device at the datacenter. Since this is a simulated environment, we will use OpenSWAN running on an EC2 host as the Customer Gateway.



Create TGW VPN Attachment

First, we need to capture the Public IP of the Customer Gateway that we will need in a later step.

1. In the EC2 Dashboard navigate to Instances
2. Select the *check box* beside the `On Premise Customer Gateway` EC2 instance and copy down the **Public IPv4 address**



Instances (1/12) Info

	Name	Instance ID	Instance state	Instance type	Status check	Alarm status
☐	VPC A Public AZ2 Server	i-0f98105b	Running	t2.micro	2/2 checks passed	No alarms
☐	VPC C Private AZ1 Server	i-00ad79a3	Running	t2.micro	2/2 checks passed	No alarms
☐	VPC A Private AZ1 Server	i-06a0a99d	Running	t2.micro	2/2 checks passed	No alarms
☐	VPC B Private AZ1 Server	i-06f60ca03	Running	t2.micro	2/2 checks passed	No alarms
☐	On-Premises DNS Server	i-05489009	Running	t3.micro	2/2 checks passed	No alarms
☐	On-Premises App Server	i-0b3a0605	Running	t3.micro	2/2 checks passed	No alarms
☑	On-Premises Customer Gateway	i-004c616b	Running	t3.micro	2/2 checks passed	No alarms

Instance: i-004c616bbb (On-Premises Customer Gateway)

Instance summary Info

Instance ID: i-004c616bl (On-Premises Customer Gateway)

Public IPv4 address: 52.

Private IPv4 addresses: 172.16.0.100

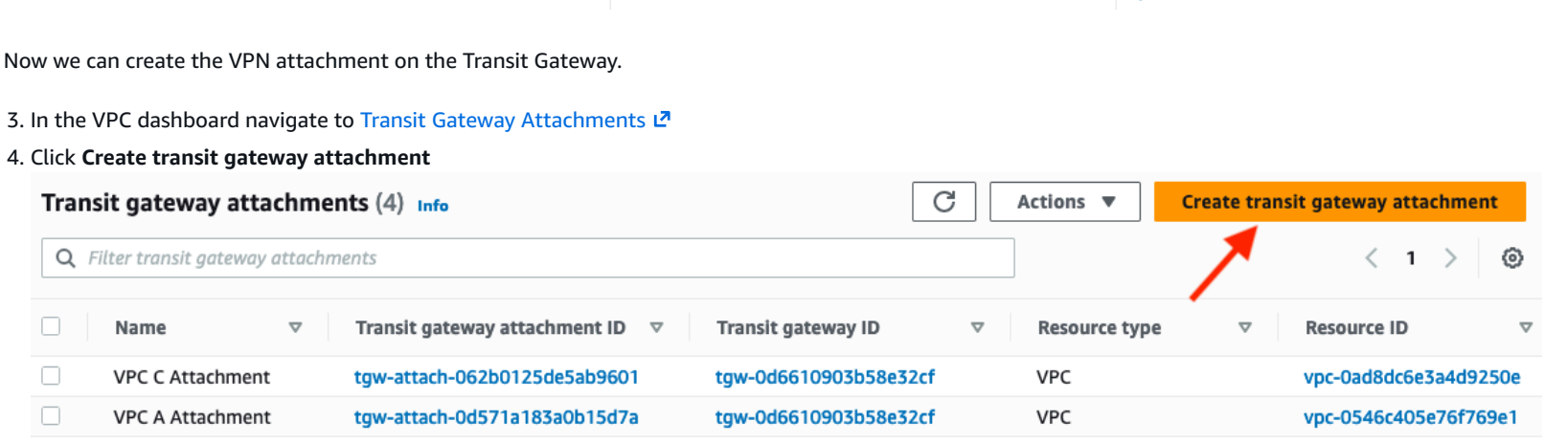
IPv6 address: –

Instance state: Running

Public IPv4 DNS: ec2-52- .compute-1.amazonaws.com

Now we can create the VPN attachment on the Transit Gateway.

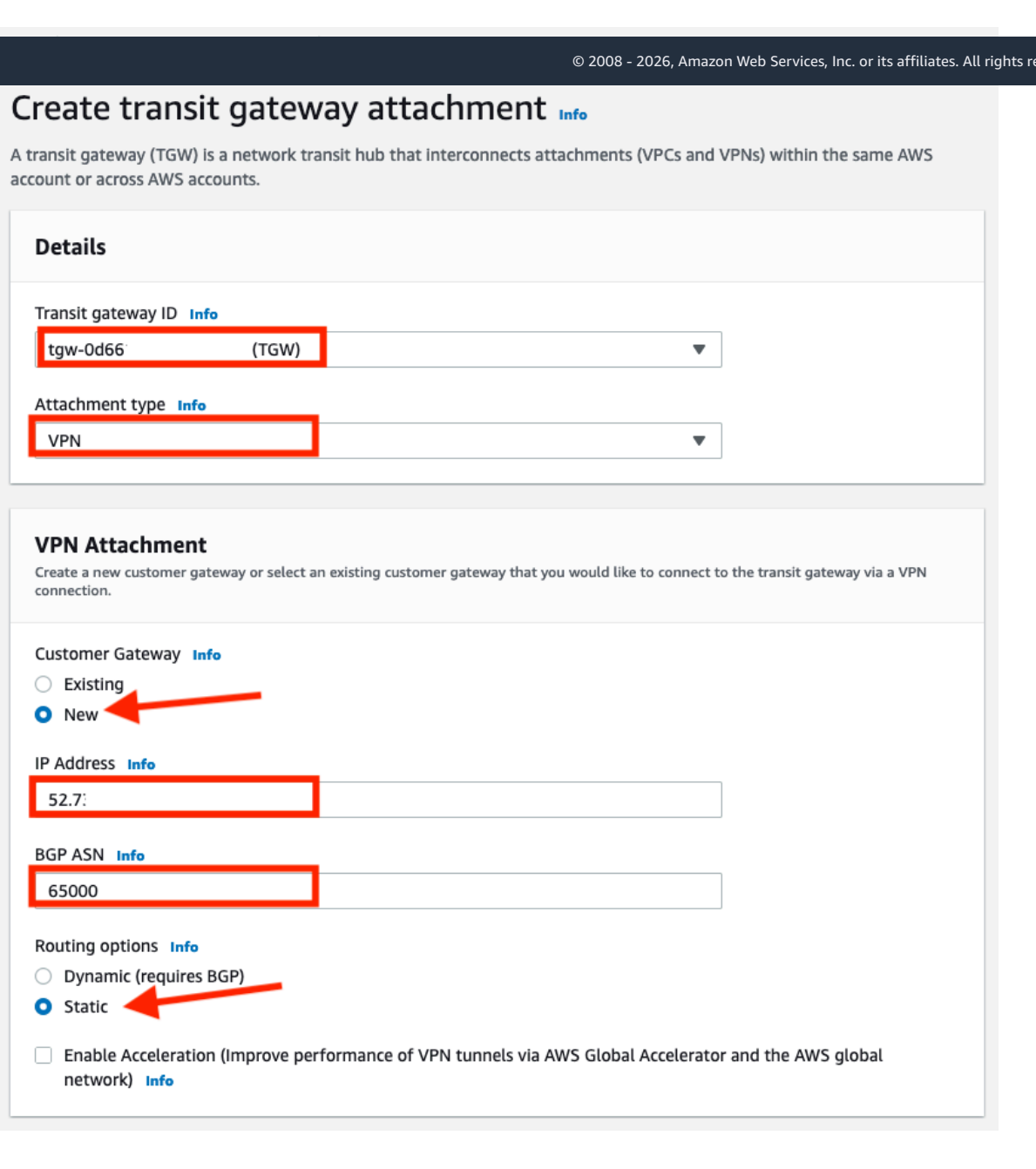
3. In the VPC dashboard navigate to Transit Gateway Attachments
4. Click **Create transit gateway attachment**


Actions ▼
Create transit gateway attachment
Filter transit gateway attachments

Transit gateway attachments (4) Info

Name	Transit gateway attachment ID	Transit gateway ID	Resource type	Resource ID
VPC C Attachment	tgw-attach-062b0125de5ab9601	tgw-0d6610903b58e32cf	VPC	vpc-0ad8dc6e3a4d9250e
VPC A Attachment	tgw-attach-0d571a183a0b15d7a	tgw-0d6610903b58e32cf	VPC	vpc-0546c405e76f769e1
VPC B Attachment	tgw-attach-0e3235c39929e7094	tgw-0d6610903b58e32cf	VPC	vpc-0d1dde426ae98bb2a

5. Under **Transit gateway ID** select the Networking Workshop TGW
6. Change **Attachment type** to **VPN**
7. Under **Customer Gateway** select **New**
8. For IP address, enter the Public IP of the `On Premise Customer Gateway` EC2 instance that you captured in a previous step.
9. Change the **Routing Options** to **Static** routing.
10. Leave all other settings at their defaults. Click **Create transit gateway attachment**



Create transit gateway attachment Info

A transit gateway (TGW) is a network transit hub that interconnects attachments (VPCs and VPNs) within the same AWS account or across AWS accounts.

Details

Transit gateway ID: tgw-0d66 (TGW)

Attachment type: VPN

VPN Attachment

Create a new customer gateway or select an existing customer gateway that you would like to connect to the transit gateway via a VPN connection.

Customer Gateway: ○ Existing ● New

IP Address: 52.7

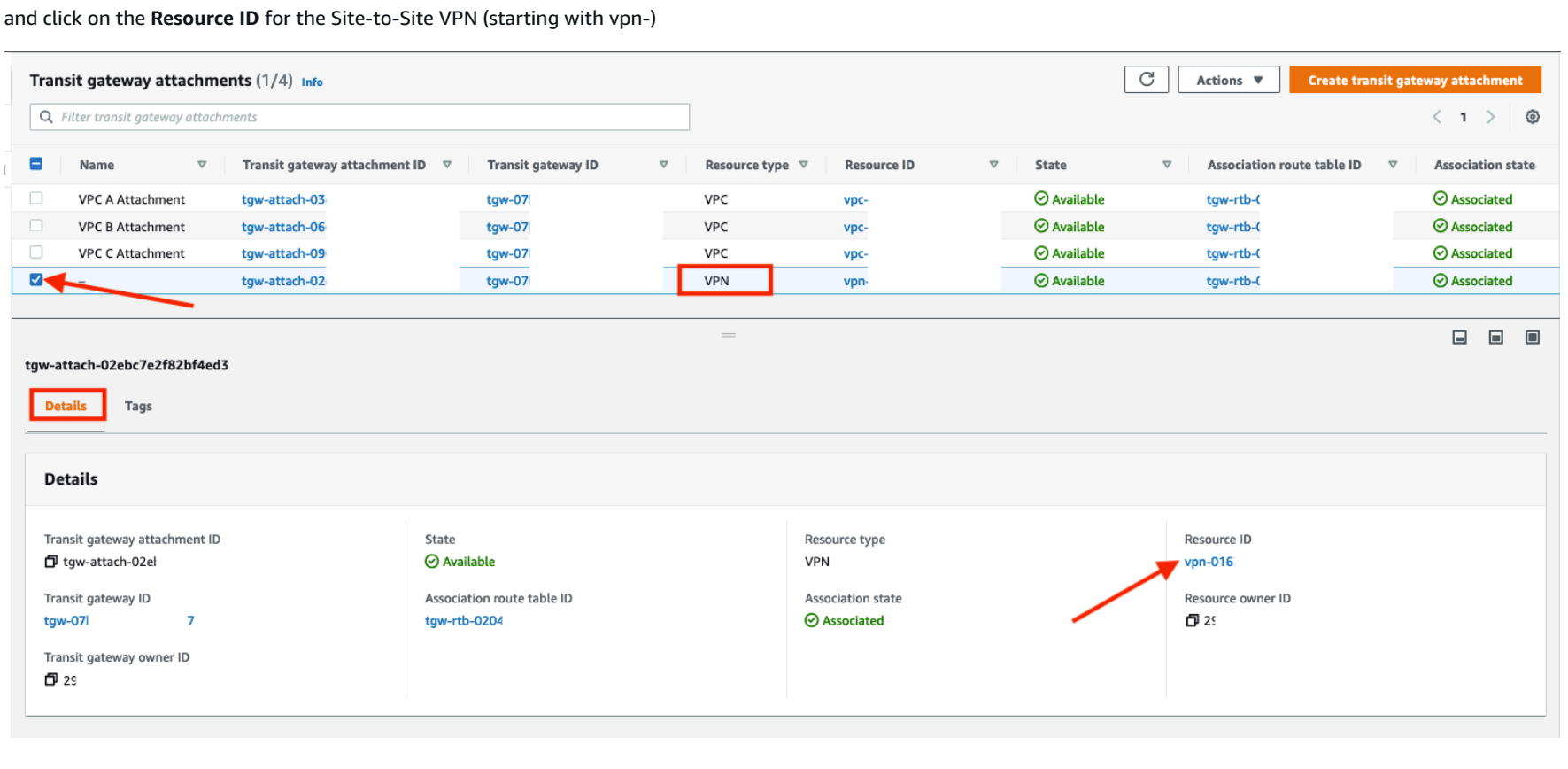
BGP ASN: 65000

Routing options: ○ Dynamic (requires BGP) ● Static

☐ Enable Acceleration (Improve performance of VPN tunnels via AWS Global Accelerator and the AWS global network)

Create Site-to-Site VPN Connection

1. Remain in the Transit Gateway Attachments dashboard and wait for the newly created VPN connection to transition to the available status. Scroll down to the details tab and click on the **Resource ID** for the Site-to-Site VPN (starting with vpn-)



Transit gateway attachments (1/4)

Name	Transit gateway attachment ID	Transit gateway ID	Resource type	Resource ID	State	Association route table ID	Association state
VPC A Attachment	tgw-attach-03	tgw-07	VPC	vpc-	Available	tgw-rtb-0	Associated
VPC B Attachment	tgw-attach-06	tgw-07	VPC	vpc-	Available	tgw-rtb-0	Associated
VPC C Attachment	tgw-attach-09	tgw-07	VPC	vpc-	Available	tgw-rtb-0	Associated
–	tgw-attach-02	tgw-07	VPN	vpn	Available	tgw-rtb-0	Associated

tgw-attach-02ebc7e2f82bf4ed3

Details | Tags

Transit gateway attachment ID: tgw-attach-02el

State: Available

Resource type: VPN

Resource ID: vpn-016

Transit gateway ID: tgw-07l

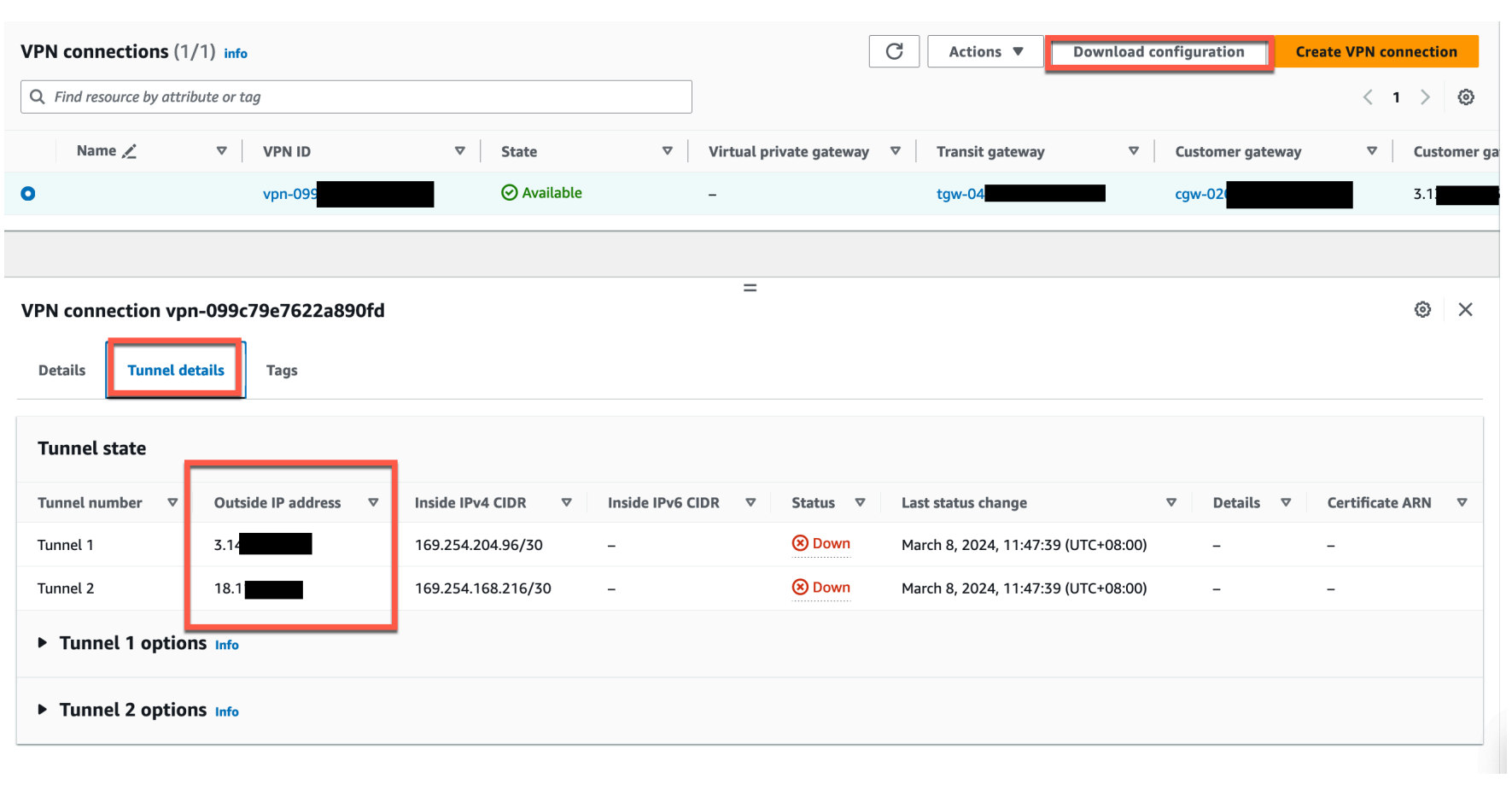
Association route table ID: tgw-rtb-0204

Association state: Associated

Resource owner ID: 2!

Transit gateway owner ID: 25

2. In the resulting screen, select the check box next to the VPN and click on the **Tunnel details** tab. Write down the two `Outside IP addresses`, you will need these at a later step. Click **Download configuration**



VPN connections (1/1) Info

Name	VPN ID	State	Virtual private gateway	Transit gateway	Customer gateway	Customer ga
	vpn-099[redacted]	Available	–	tgw-04[redacted]	cgw-02[redacted]	3.1[redacted]

VPN connection vpn-099c79e7622a890fd

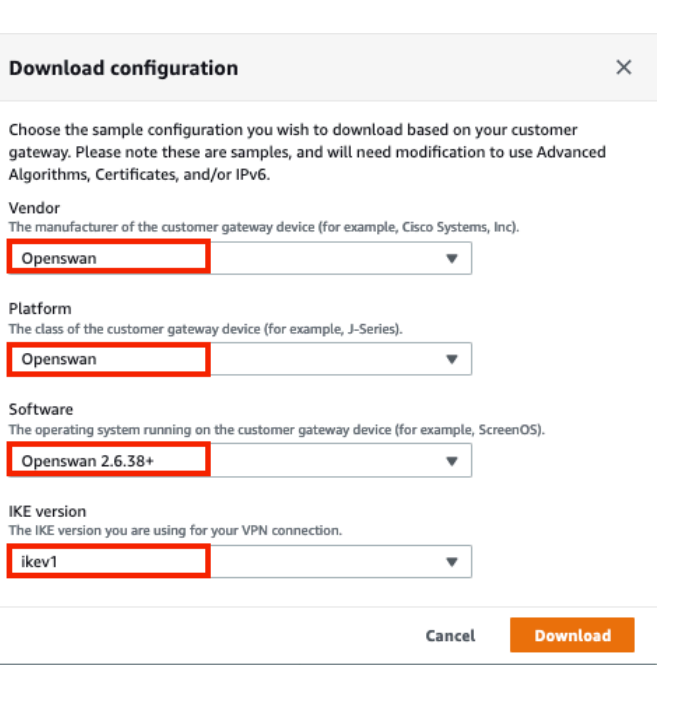
Tunnel state

Tunnel number	Outside IP address	Inside IPv4 CIDR	Inside IPv6 CIDR	Status	Last status change	Details	Certificate ARN
Tunnel 1	3.14[redacted]	169.254.204.96/30	–	Down	March 8, 2024, 11:47:39 (UTC+08:00)	–	–
Tunnel 2	18.1[redacted]	169.254.168.216/30	–	Down	March 8, 2024, 11:47:39 (UTC+08:00)	–	–

▸ Tunnel 1 options Info

▸ Tunnel 2 options Info

3. Select `Openswan` for the vendor.



Download configuration

Choose the sample configuration you wish to download based on your customer gateway. Please note these are samples, and will need modification to use Advanced Algorithms, Certificates, and/or IPv6.

Vendor: Openswan

Platform: Openswan

Software: Openswan 2.6.38+

IKE version: ikev1

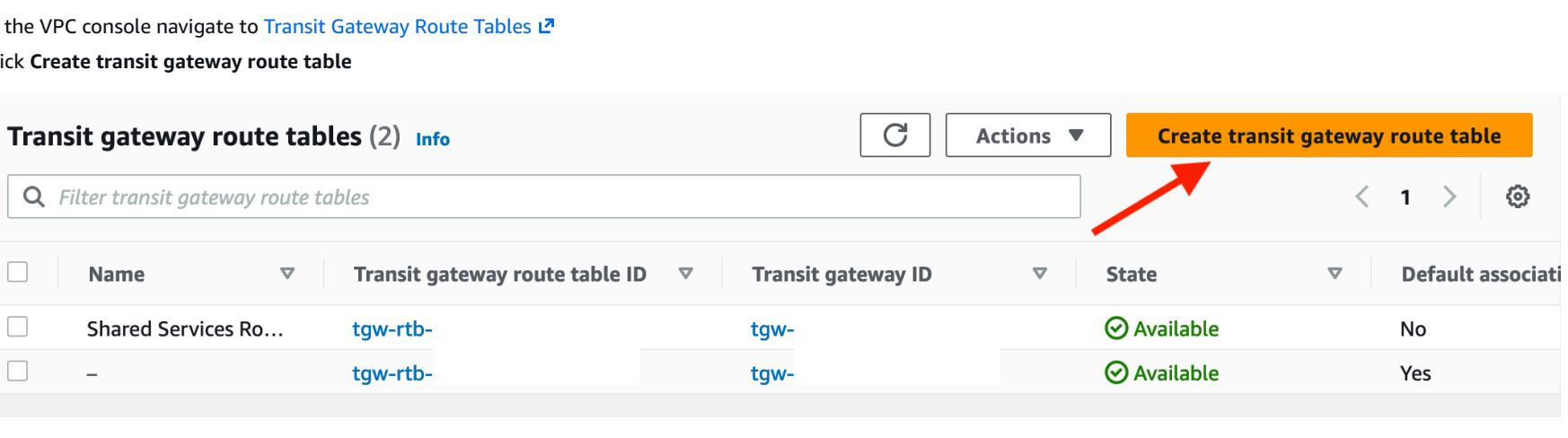
Cancel | Download

4. Click **Download** and save the downloaded file for later.
5. Click **Cancel** to close the window.

Create a new Transit Gateway Route Table for the VPN

The new VPN connection needs to be associated to a transit gateway route table. Since none of the existing ones are suitable (they do not have all the VPCs routes), we will create a new one.

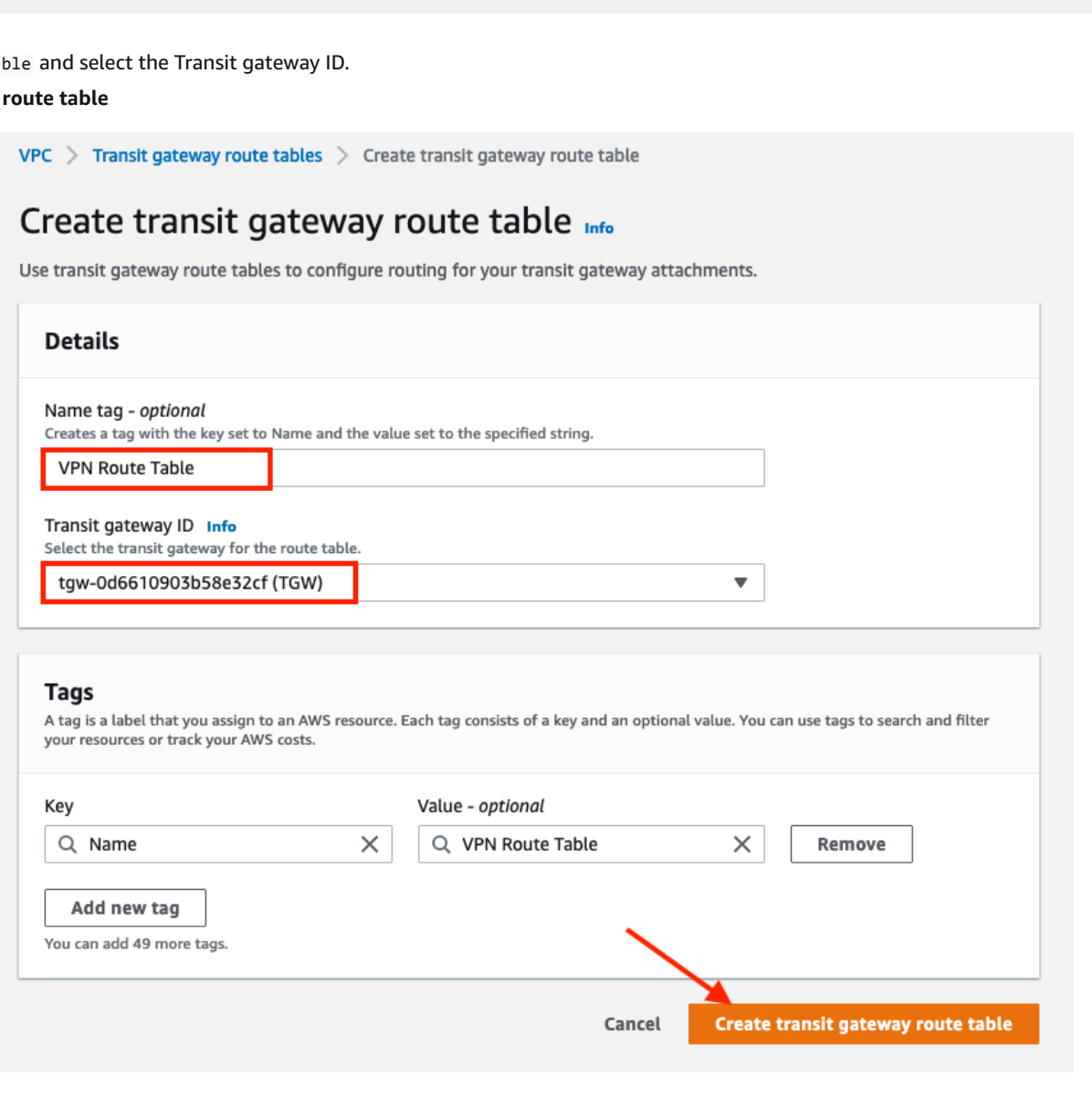
1. In the VPC console navigate to Transit Gateway Route Tables
2. Click **Create transit gateway route table**



Transit gateway route tables (2) Info

Name	Transit gateway route table ID	Transit gateway ID	State	Default associati
Shared Services Ro...	tgw-rtb-	tgw-	Available	No
–	tgw-rtb-	tgw-	Available	Yes

3. Enter the name `VPN Route Table` and select the Transit gateway ID.
4. Click **Create transit gateway route table**



VPC > Transit gateway route tables > Create transit gateway route table

Create transit gateway route table Info

Use transit gateway route tables to configure routing for your transit gateway attachments.

Details

Name tag - *optional*: VPN Route Table

Transit gateway ID: tgw-0d6610903b58e32cf (TGW)

Tags

Key: Name | Value - *optional*: VPN Route Table | Remove

Add new tag

You can add 49 more tags.

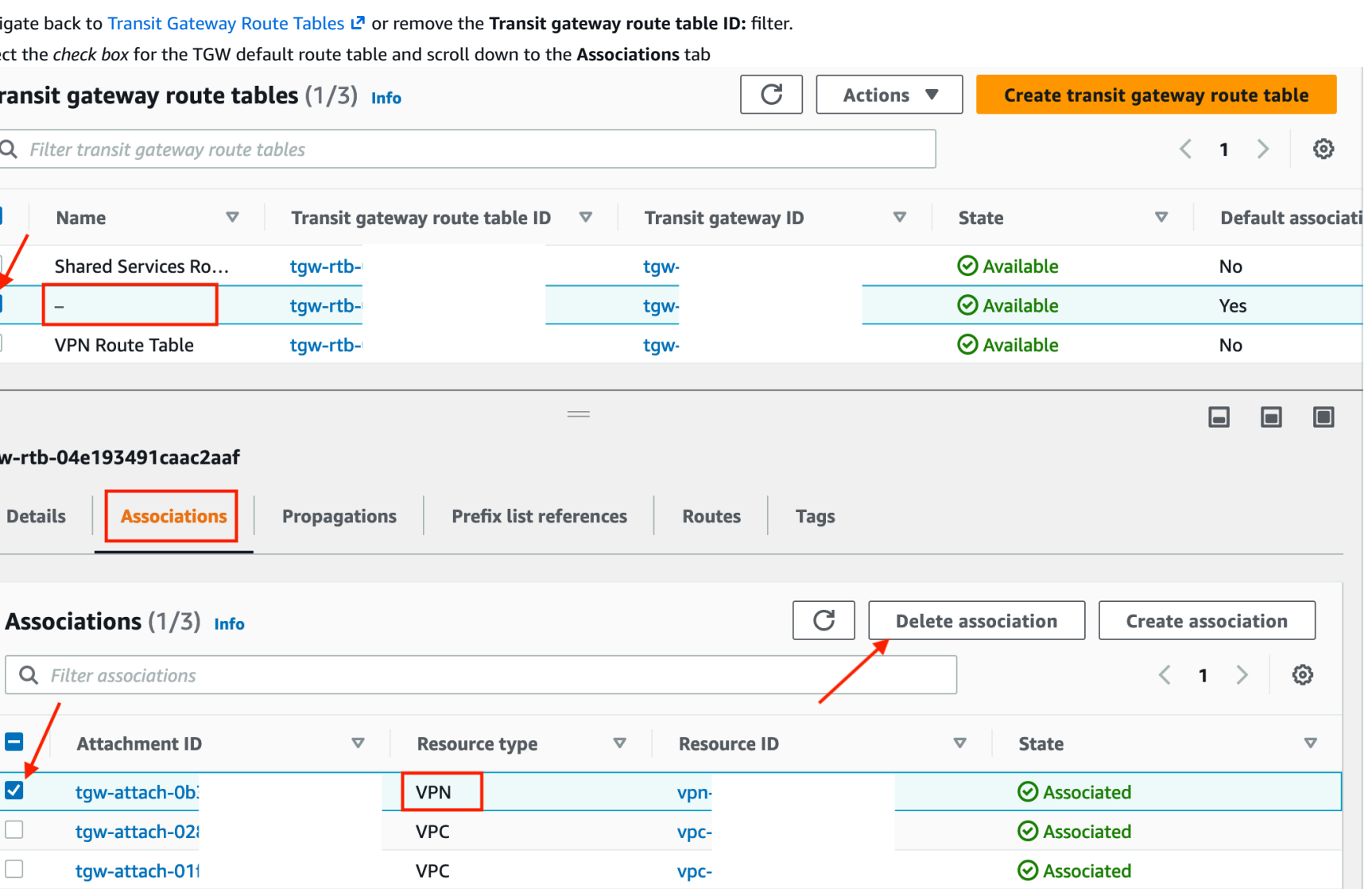
Cancel | Create transit gateway route table

Delete VPN Attachment from Transit Gateway default Route Table

When we created the transit gateway, we left the option `Default route table association` enabled. This means that when we created the VPN attachment, it was automatically associated with the transit gateways default route table. Before we associate the VPN attachment to the new VPN route table, we have to disassociate it from the default route table.

> ⓘ If you started this workshop after the **Multiple VPCs** section, you do not need to complete this step. `Default route table association` **is not** enabled in the `NetworkingWorkshopMultiVPCandTGW` CloudFormation template deployed in the Security Controls or Connecting to On-Premises prerequisites sections.

1. Navigate back to Transit Gateway Route Tables or remove the **Transit gateway route table ID:** filter.
2. Select the *check box* for the TGW default route table and scroll down to the **Associations** tab



Transit gateway route tables (1/3) Info

	Name	Transit gateway route table ID	Transit gateway ID	State	Default associati
☐	Shared Services Ro...	tgw-rtb-	tgw-	Available	No
☑	–	tgw-rtb-	tgw-	Available	Yes
☐	VPN Route Table	tgw-rtb-	tgw-	Available	No

tgw-rtb-04e193491caac2aaf

Details | Associations | Propagations | Prefix list references | Routes | Tags

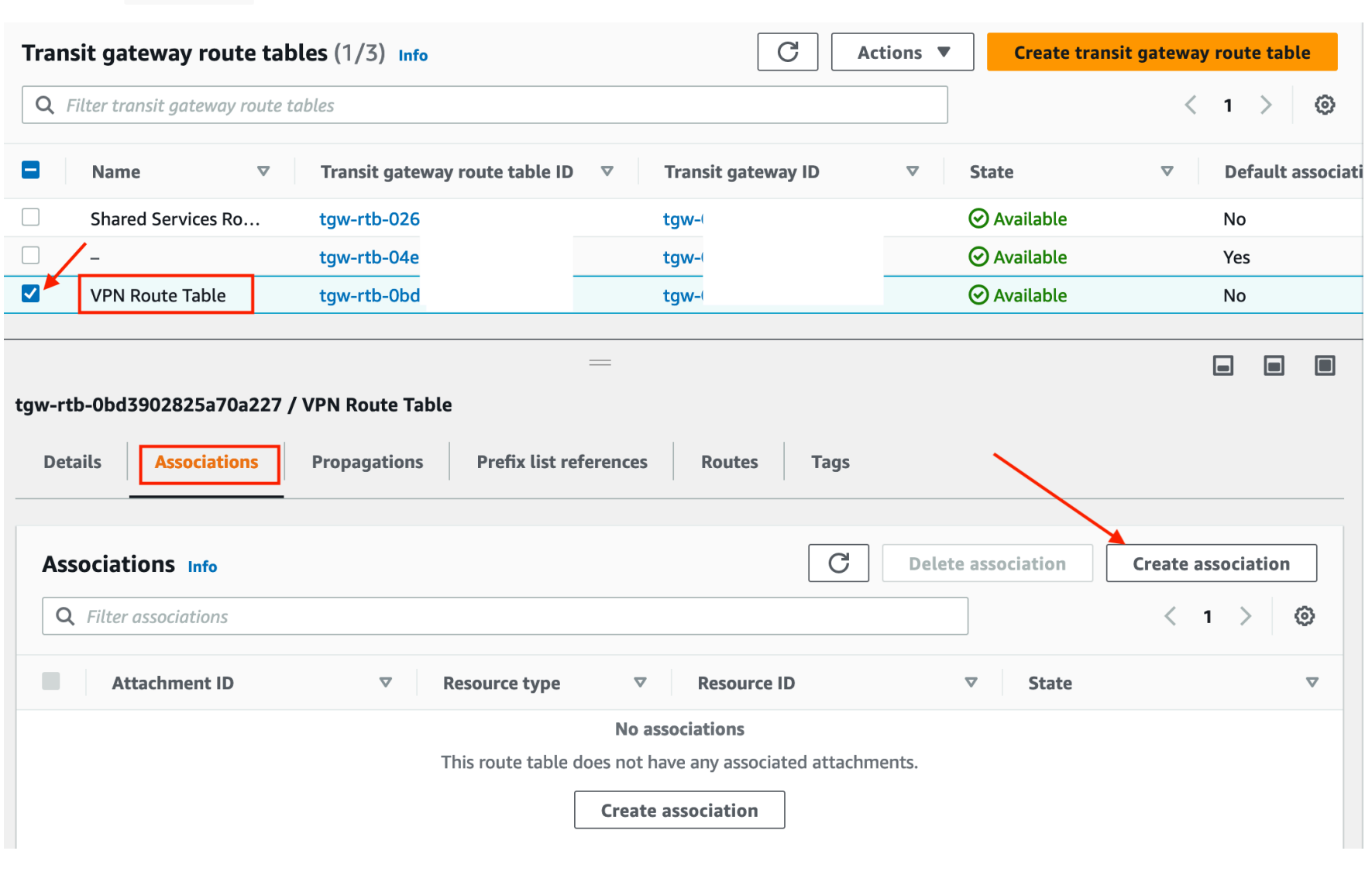
Associations (1/3) Info — Delete association | Create association

	Attachment ID	Resource type	Resource ID	State
☑	tgw-attach-0b	VPN	vpn-	Associated
☐	tgw-attach-02	VPC	vpc-	Associated
☐	tgw-attach-01	VPC	vpc-	Associated

3. Select the association with the resource type **VPN** and click **Delete association**
4. Confirm the deletion by clicking **Delete Association** on the following screen
5. The association will move into *disassociating* state.

Associate the VPN attachment to the new Transit Gateway Route Table

1. Select the new `VPN Route Table` and click **Associations** tab. Click **Create association**



Transit gateway route tables (1/3) Info

	Name	Transit gateway route table ID	Transit gateway ID	State	Default associati
☐	Shared Services Ro...	tgw-rtb-026	tgw-	Available	No
☐	–	tgw-rtb-04e	tgw-	Available	Yes
☑	VPN Route Table	tgw-rtb-0bd	tgw-	Available	No

tgw-rtb-0bd3902825a70a227 / VPN Route Table

Details | Associations | Propagations | Prefix list references | Routes | Tags

Associations Info — Delete association | Create association

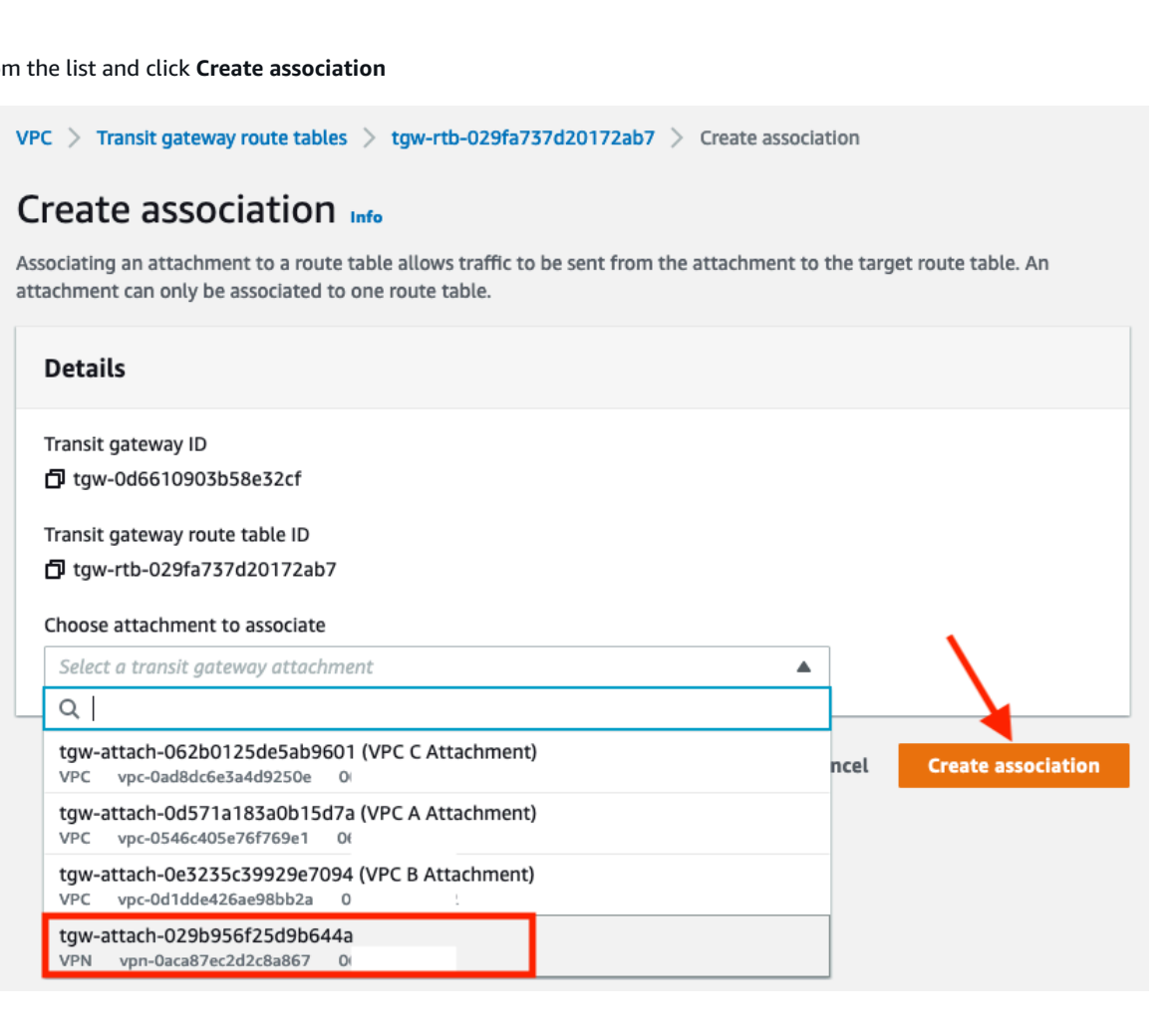
Attachment ID	Resource type	Resource ID	State

No associations

This route table does not have any associated attachments.

Create association

2. Select the VPN attachment from the list and click **Create association**



VPC > Transit gateway route tables > tgw-rtb-029fa737d20172ab7 > Create association

Create association Info

Associating an attachment to a route table allows traffic to be sent from the attachment to the target route table. An attachment can only be associated to one route table.

Details

Transit gateway ID: tgw-0d6610903b58e32cf

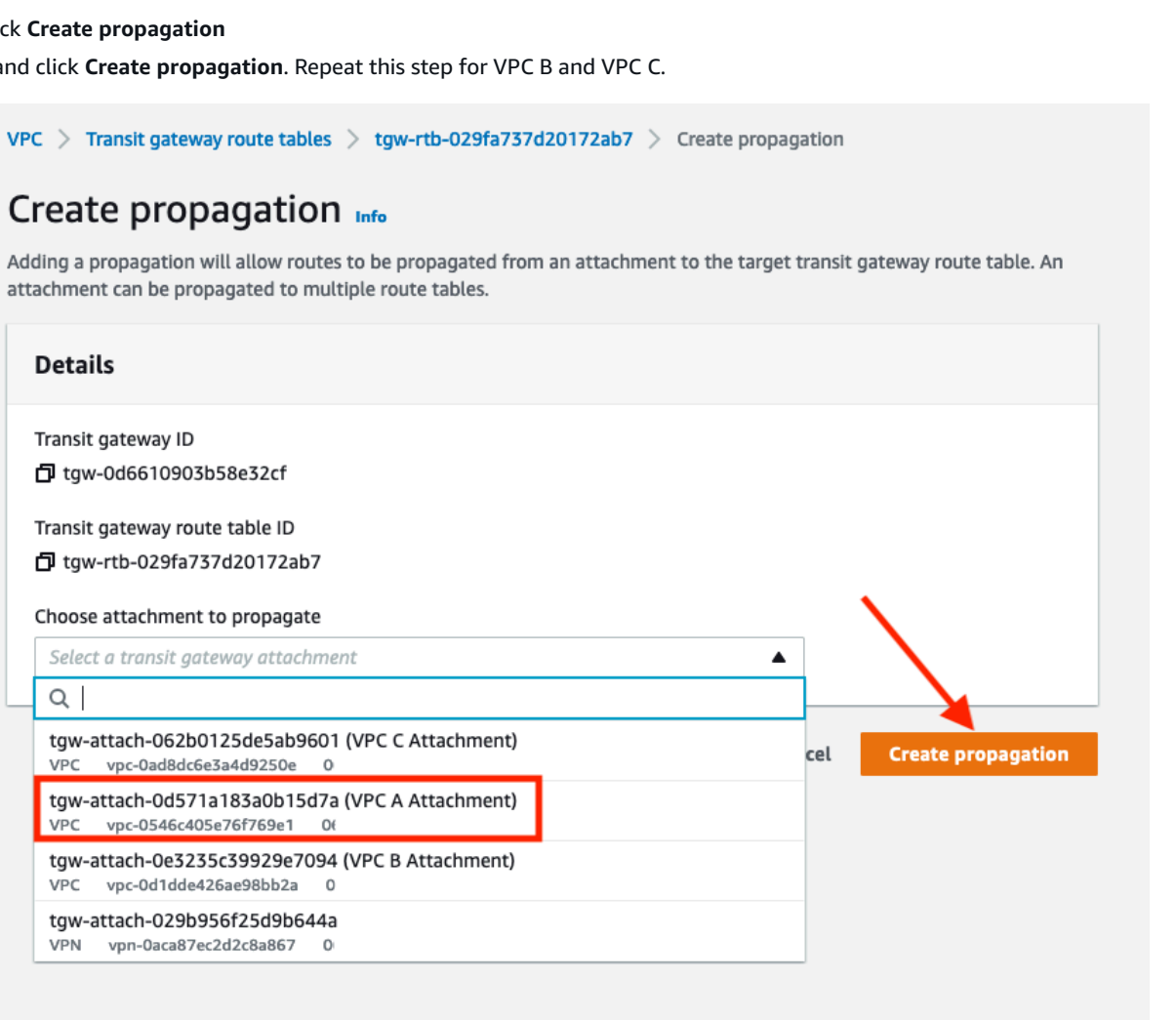
Transit gateway route table ID: tgw-rtb-029fa737d20172ab7

Choose attachment to associate: Select a transit gateway attachment

- tgw-attach-062b0125de5ab9601 (VPC C Attachment) VPC vpc-0ad8dc6e3a4d9250e
- tgw-attach-0d571a183a0b15d7a (VPC A Attachment) VPC vpc-0546c405e76f769e1
- tgw-attach-0e3235c39929e7094 (VPC B Attachment) VPC vpc-0d1dde426ae98bb2a
- tgw-attach-029b956f25d9b644a VPN vpn-0aca87ec2d2c8a867

Create association

3. Click **Propagations** tab and click **Create propagation**
4. Select the VPC A attachment and click **Create propagation**. Repeat this step for VPC B and VPC C.



VPC > Transit gateway route tables > tgw-rtb-029fa737d20172ab7 > Create propagation

Create propagation Info

Adding a propagation will allow routes to be propagated from an attachment to the target transit gateway route table. An attachment can be propagated to multiple route tables.

Details

Transit gateway ID: tgw-0d6610903b58e32cf

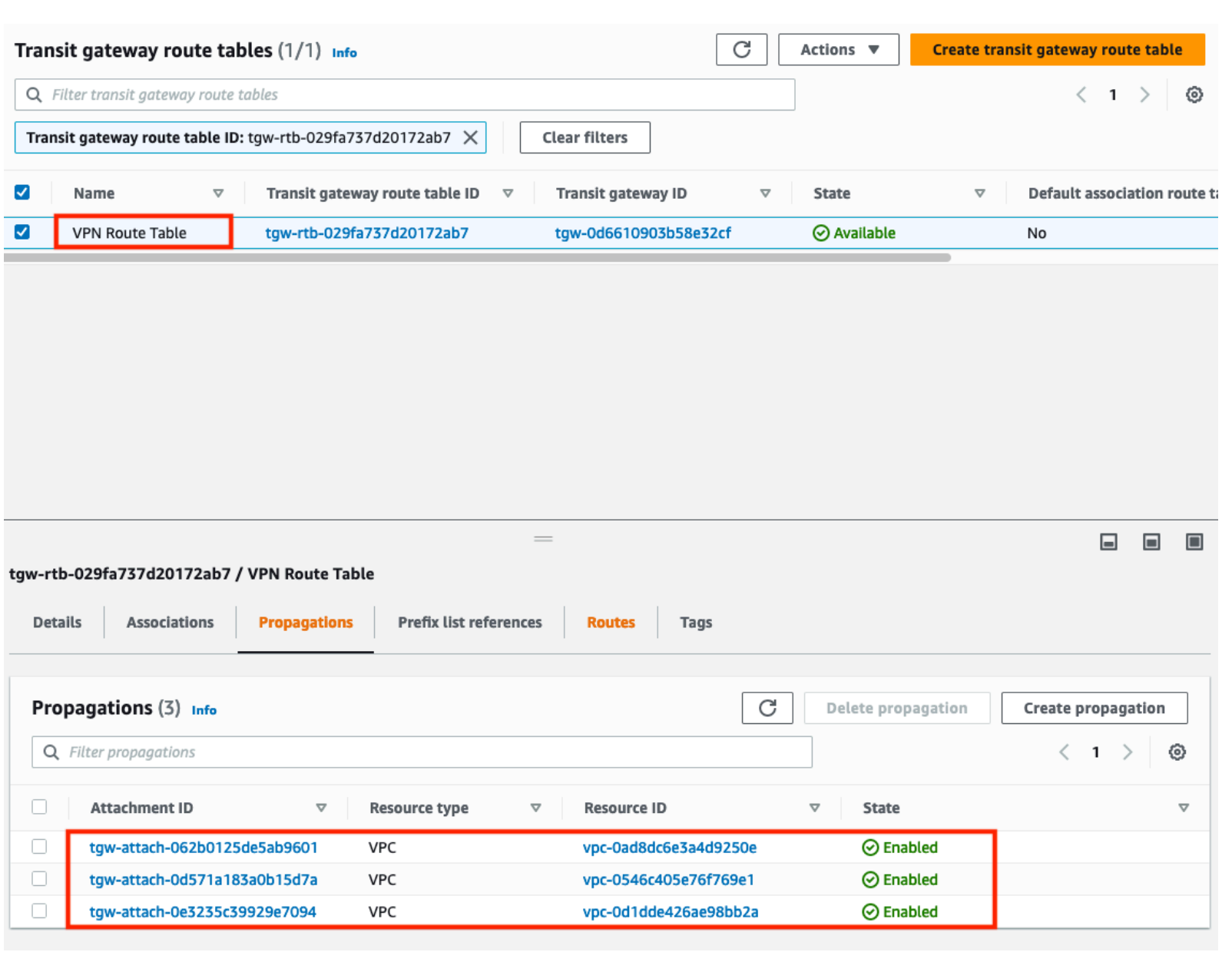
Transit gateway route table ID: tgw-rtb-029fa737d20172ab7

Choose attachment to propagate: Select a transit gateway attachment

- tgw-attach-062b0125de5ab9601 (VPC C Attachment) VPC vpc-0ad8dc6e3a4d9250e
- tgw-attach-0d571a183a0b15d7a (VPC A Attachment) VPC vpc-0546c405e76f769e1
- tgw-attach-0e3235c39929e7094 (VPC B Attachment) VPC vpc-0d1dde426ae98bb2a
- tgw-attach-029b956f25d9b644a VPN vpn-0aca87ec2d2c8a867

Create propagation

The VPN connection is now associated to this route table, and has the ability to route to VPC A, B, and C.



Transit gateway route tables (1/1) Info

Transit gateway route table ID: tgw-rtb-029fa737d20172ab7 | Clear filters

	Name	Transit gateway route table ID	Transit gateway ID	State	Default association route t
☑	VPN Route Table	tgw-rtb-029fa737d20172ab7	tgw-0d6610903b58e32cf	Available	No

tgw-rtb-029fa737d20172ab7 / VPN Route Table

Details | Associations | Propagations | Prefix list references | Routes | Tags

Propagations (3) Info — Delete propagation | Create propagation

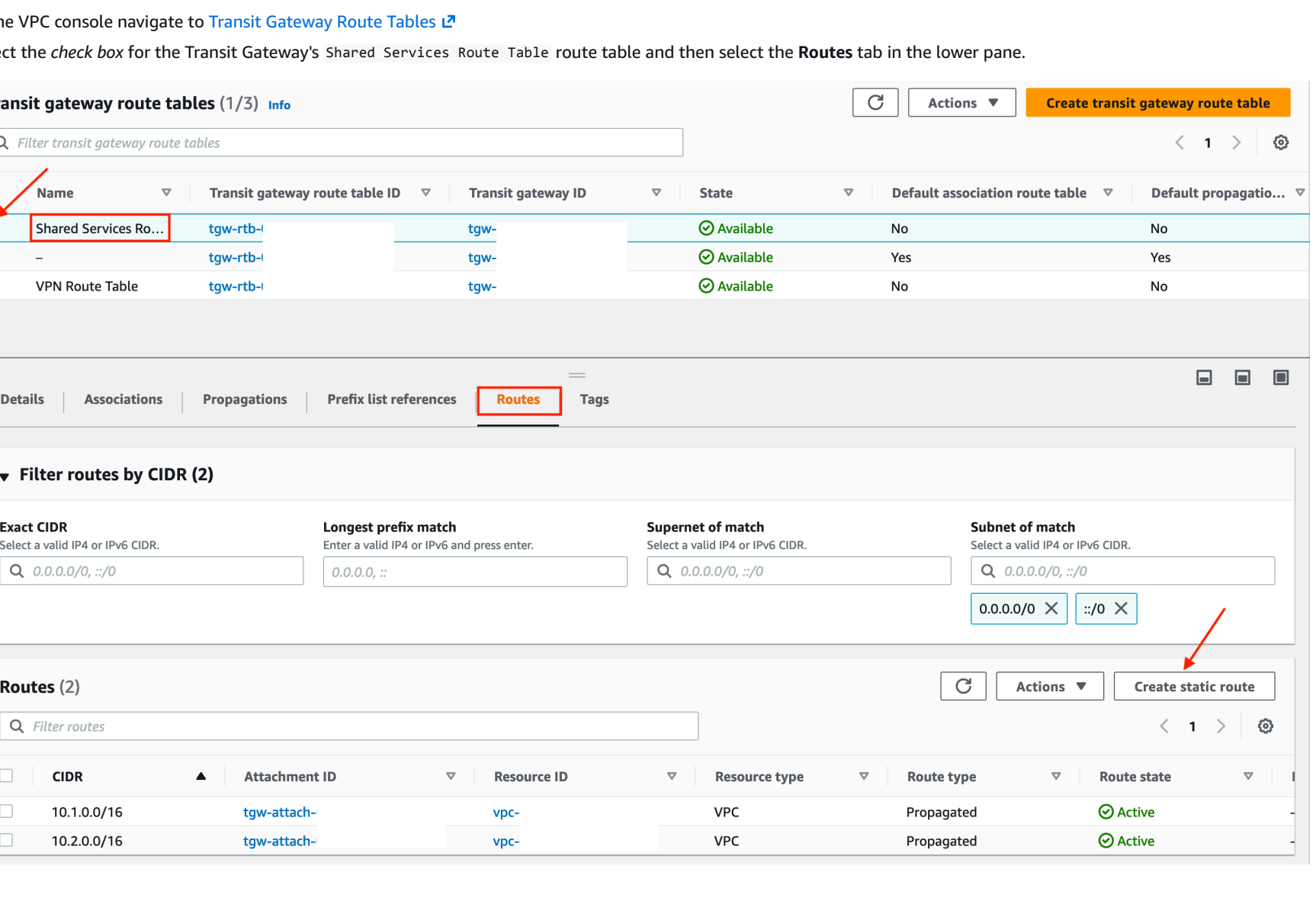
Attachment ID	Resource type	Resource ID	State
tgw-attach-062b0125de5ab9601	VPC	vpc-0ad8dc6e3a4d9250e	Enabled
tgw-attach-0d571a183a0b15d7a	VPC	vpc-0546c405e76f769e1	Enabled
tgw-attach-0e3235c39929e7094	VPC	vpc-0d1dde426ae98bb2a	Enabled

Update Route Tables with On-Premise CIDR

Update Transit Gateway Route Table

Since we are using static routing for our VPN connection, we need to manually create a route in the Transit Gateway route table to route traffic to the on-premise CIDR block via the new VPN attachment.

1. In the VPC console navigate to Transit Gateway Route Tables
2. Select the *check box* for the Transit Gateway's `Shared Services Route Table` route table and then select the **Routes** tab in the lower pane.



Transit gateway route tables (1/3) Info

	Name	Transit gateway route table ID	Transit gateway ID	State	Default association route table	Default propagatio...
☑	Shared Services Ro...	tgw-rtb-	tgw-	Available	No	No
☐	–	tgw-rtb-	tgw-	Available	Yes	Yes
☐	VPN Route Table	tgw-rtb-	tgw-	Available	No	No

Details | Associations | Propagations | Prefix list references | Routes | Tags

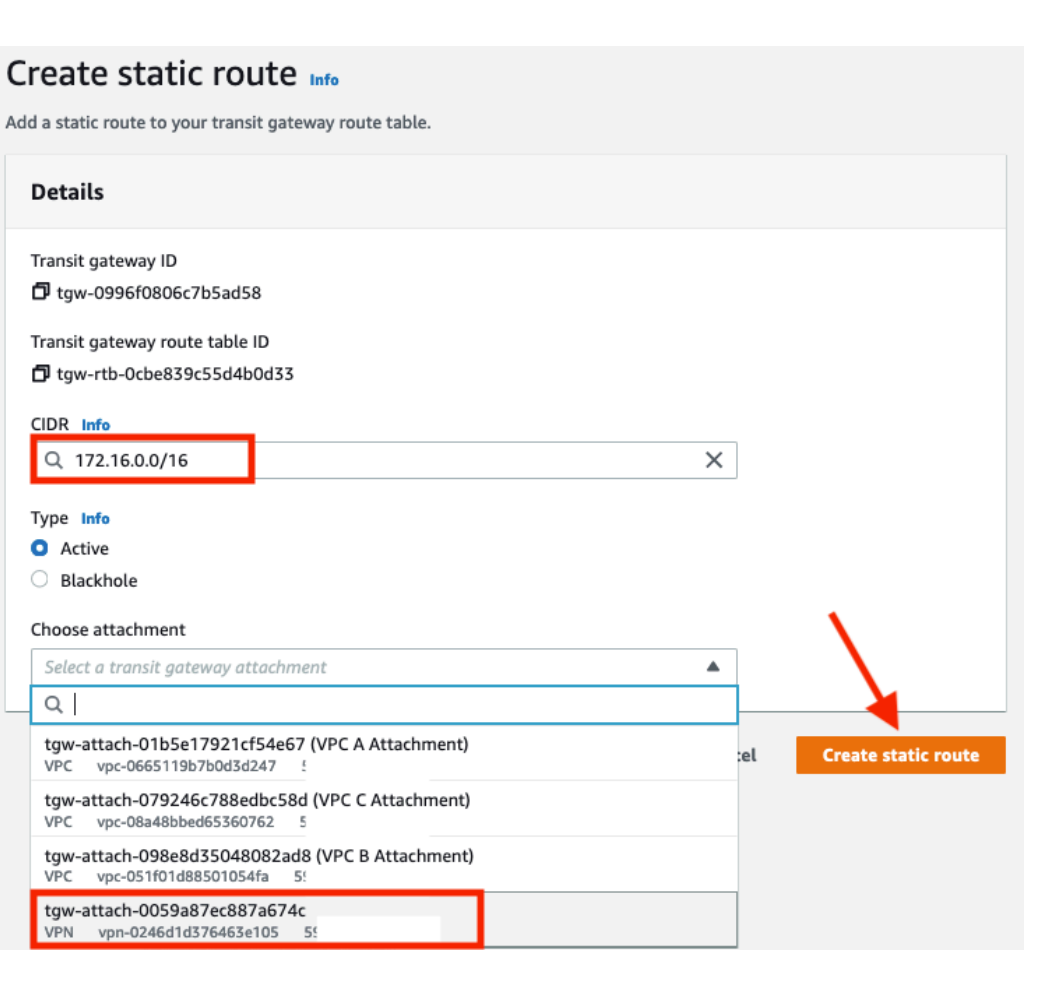
▾ Filter routes by CIDR (2)

Exact CIDR | Longest prefix match | Supernet of match | Subnet of match: 0.0.0.0/0 ✕ ::/0 ✕

Routes (2) — Actions ▼ | Create static route

CIDR	Attachment ID	Resource ID	Resource type	Route type	Route state
10.1.0.0/16	tgw-attach-	vpc-	VPC	Propagated	Active
10.2.0.0/16	tgw-attach-	vpc-	VPC	Propagated	Active

3. Click **Create static route** to add a new static route.
4. Enter the CIDR block for the simulated data center environment `172.16.0.0/16` and select the new VPN attachment. There should be 3 attachments of type VPC and one of type VPN.
5. Click **Create static route**



Create static route Info

Add a static route to your transit gateway route table.

Details

Transit gateway ID: tgw-0996f0806c7b5ad58

Transit gateway route table ID: tgw-rtb-0cbe839c55d4b0d33

CIDR: 172.16.0.0/16

Type: ● Active ○ Blackhole

Choose attachment: Select a transit gateway attachment

- tgw-attach-01b5e17921cf54e67 (VPC A Attachment) VPC vpc-0665119b7b0d3d247
- tgw-attach-079246c788edbc58d (VPC C Attachment) VPC vpc-08a48bbed65360762
- tgw-attach-098e8d35048082ad8 (VPC B Attachment) VPC vpc-051f01d88501054fa
- tgw-attach-0059a87ec887a674c VPN vpn-0246d1d376463e105

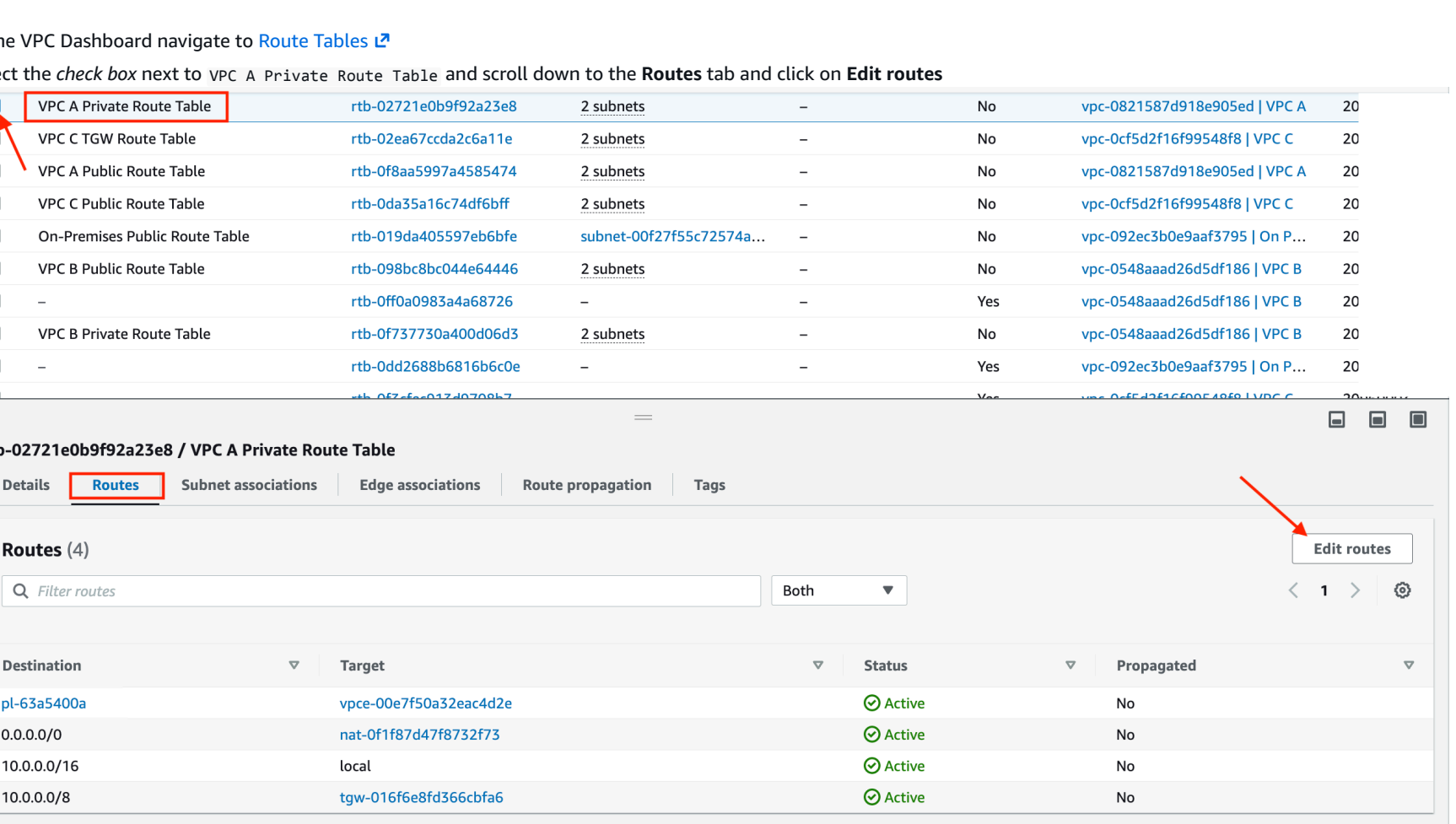
Create static route

6. It will initially be created with a route state of 'Blackhole' because we have yet to establish the VPN connection to the on-premise environment.
7. Repeat these steps to update the `Default Route Table` so that it also has a route to the simulated data center environment.

Update VPC A Route Table

Now that we have a Transit Gateway Attachment through which to send traffic to our on-premises network we need to add routes from our AWS VPCs to the Transit Gateway for the on-premise CIDR block. For the purposes of this lab you're only going to add an entry to the private route table for VPC A.

1. In the VPC Dashboard navigate to Route Tables
2. Select the *check box* next to `VPC A Private Route Table` and scroll down to the **Routes** tab and click on **Edit routes**



☑	VPC A Private Route Table	rtb-02721e0b9f92a23e8	2 subnets	–	No	vpc-0821587d918e905ed | VPC A	20
☐	VPC C TGW Route Table	rtb-02ea67ccda2c6a11e	2 subnets	–	No	vpc-0cf5d2f16f99548f8 | VPC C	20
☐	VPC A Public Route Table	rtb-0f8aa5997a4585474	2 subnets	–	No	vpc-0821587d918e905ed | VPC A	20
☐	VPC C Public Route Table	rtb-0da35a16c74df6bff	2 subnets	–	No	vpc-0cf5d2f16f99548f8 | VPC C	20
☐	On-Premises Public Route Table	rtb-019da405597eb6bfe	subnet-00f27f55c72574a...	–	No	vpc-092ec3b0e9aaf3795 | On P...	20
☐	VPC B Public Route Table	rtb-098bc8bc044e64446	2 subnets	–	No	vpc-0548aaad26d5df186 | VPC B	20
☐	–	rtb-0ff0a0983a4a68726	–	–	Yes	vpc-0548aaad26d5df186 | VPC B	20
☐	VPC B Private Route Table	rtb-0f737730a400d06d3	2 subnets	–	No	vpc-0548aaad26d5df186 | VPC B	20
☐	–	rtb-0dd2688b6816b6c0e	–	–	Yes	vpc-092ec3b0e9aaf3795 | On P...	20

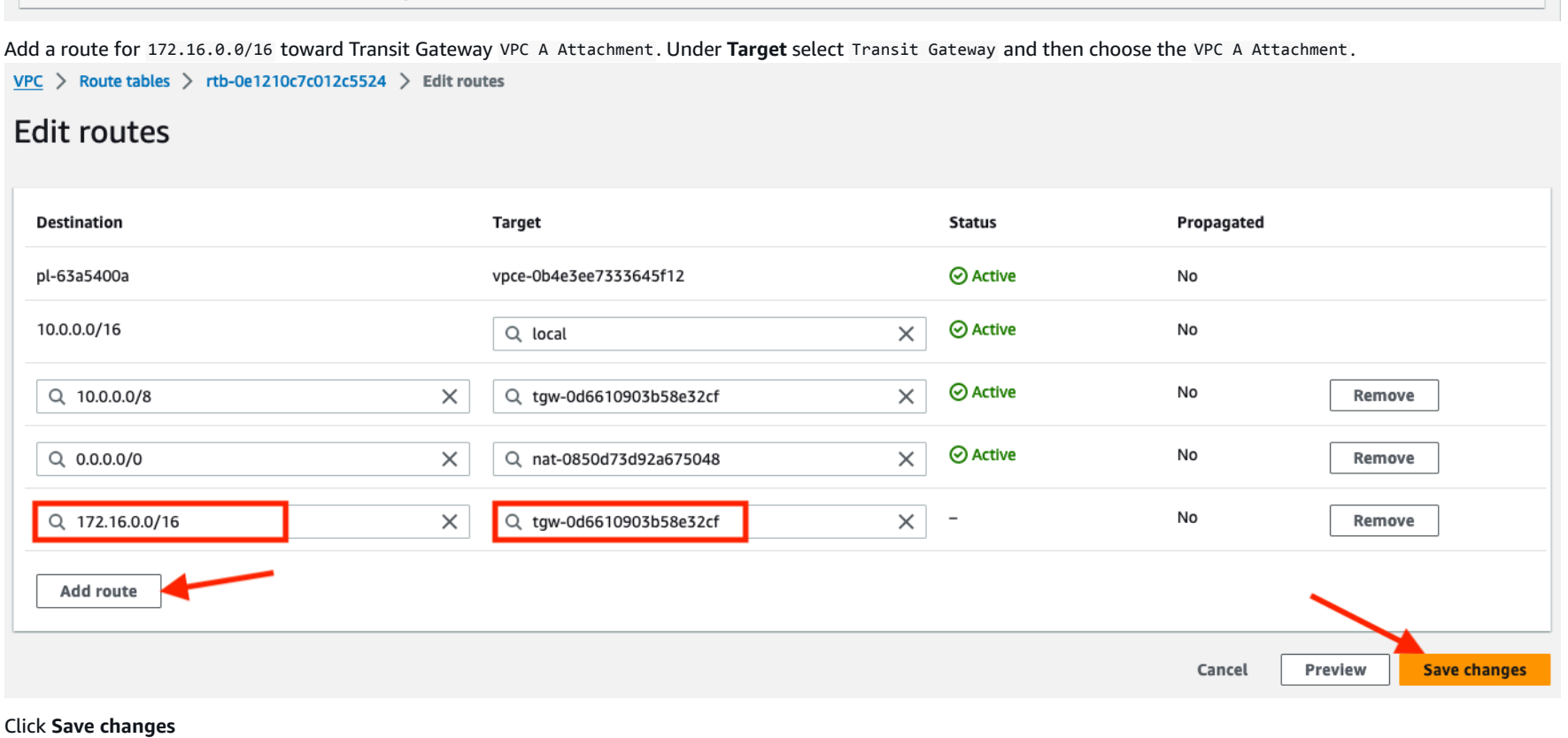
rtb-02721e0b9f92a23e8 / VPC A Private Route Table

Details | Routes | Subnet associations | Edge associations | Route propagation | Tags

Routes (4) — Edit routes

Destination	Target	Status	Propagated
pl-63a5400a	vpce-00e7f50a32eac4d2e	Active	No
0.0.0.0/0	nat-0f1f87d47f8732f73	Active	No
10.0.0.0/16	local	Active	No
10.0.0.0/8	tgw-016f6e8fd366cbfa6	Active	No

3. Add a route for `172.16.0.0/16` toward Transit Gateway `VPC A Attachment`. Under **Target** select `Transit Gateway` and then choose the `VPC A Attachment`.



VPC > Route tables > rtb-0e1210c7c012c5524 > Edit routes

Edit routes

Destination	Target	Status	Propagated	
pl-63a5400a	vpce-0b4e3ee7333645f12	Active	No	
10.0.0.0/16	local	Active	No	
10.0.0.0/8	tgw-0d6610903b58e32cf	Active	No	Remove
0.0.0.0/0	nat-0850d73d92a675048	Active	No	Remove
172.16.0.0/16	tgw-0d6610903b58e32cf	–	No	Remove

Add route

Cancel | Preview | Save changes

4. Click **Save changes**

VPC A will now have the ability to route to the simulated on-premesis environment

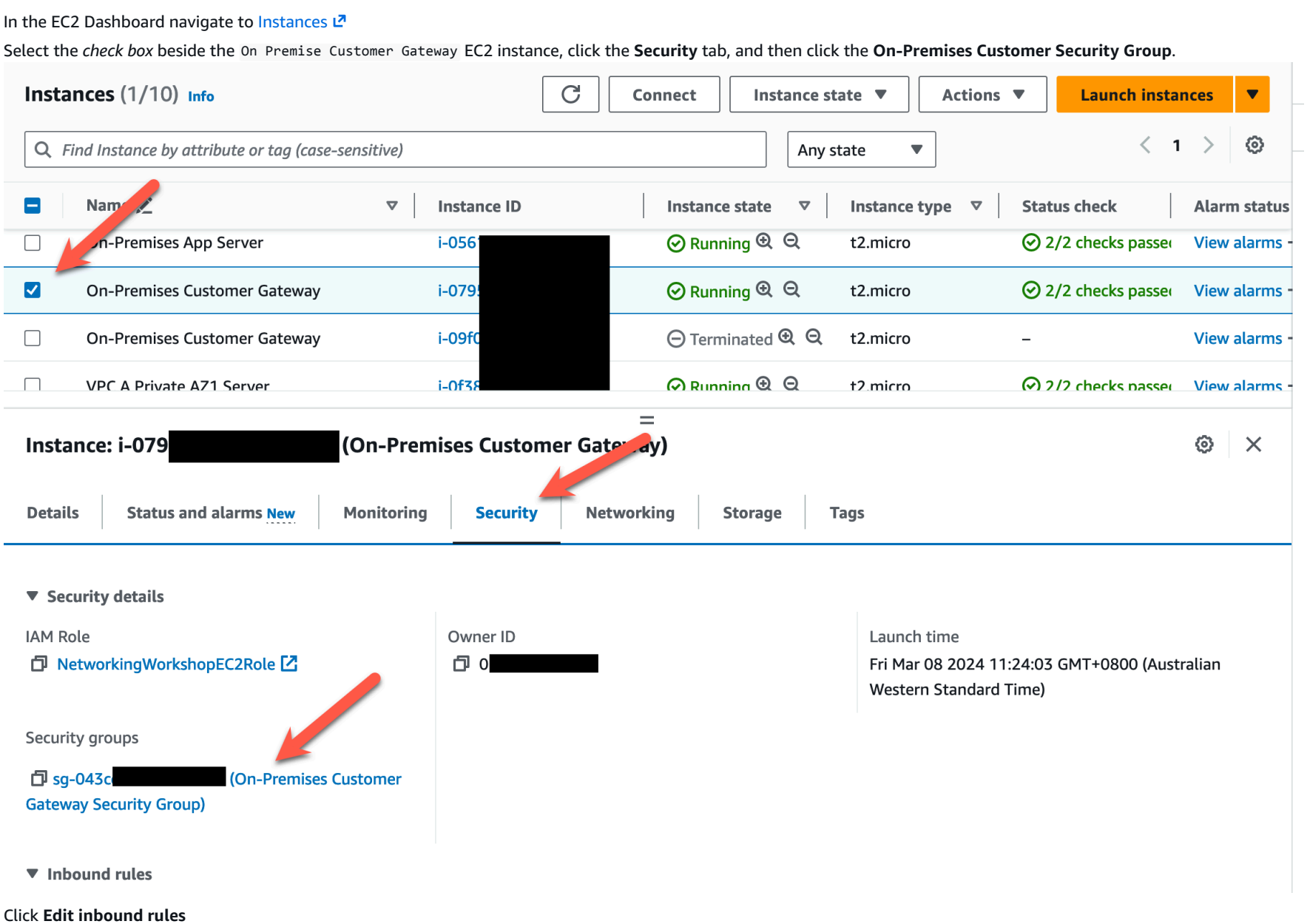
Please repeat this step for the VPC B and VPC C Private Route Tables.

There is no need to configure the routing in the on-premesis VPC. This was configured for you in the on-premesis CloudFormation template.

Update the customer gateway EC2 instance's security group

We will need to allow the AWS VPN endpoints to communicate with the `On Premise Customer Gateway` EC2 instance over IPSEC.

1. In the EC2 Dashboard navigate to Instances
2. Select the *check box* beside the `On Premise Customer Gateway` EC2 instance, click the **Security** tab, and then click the **On-Premises Customer Security Group**.



Instances (1/10) Info

	Name	Instance ID	Instance state	Instance type	Status check	Alarm status
☐	On-Premises App Server	i-056[redacted]	Running	t2.micro	2/2 checks passed	View alarms
☑	On-Premises Customer Gateway	i-079[redacted]	Running	t2.micro	2/2 checks passed	View alarms
☐	On-Premises Customer Gateway	i-09f0[redacted]	Terminated	t2.micro	–	View alarms
☐	VPC A Private AZ1 Server	i-0f3[redacted]	Running	t2.micro	2/2 checks passed	View alarms

Instance: i-079[redacted] (On-Premises Customer Gateway)

Details | Status and alarms New | Monitoring | Security | Networking | Storage | Tags

▾ Security details

IAM Role: NetworkingWorkshopEC2Role

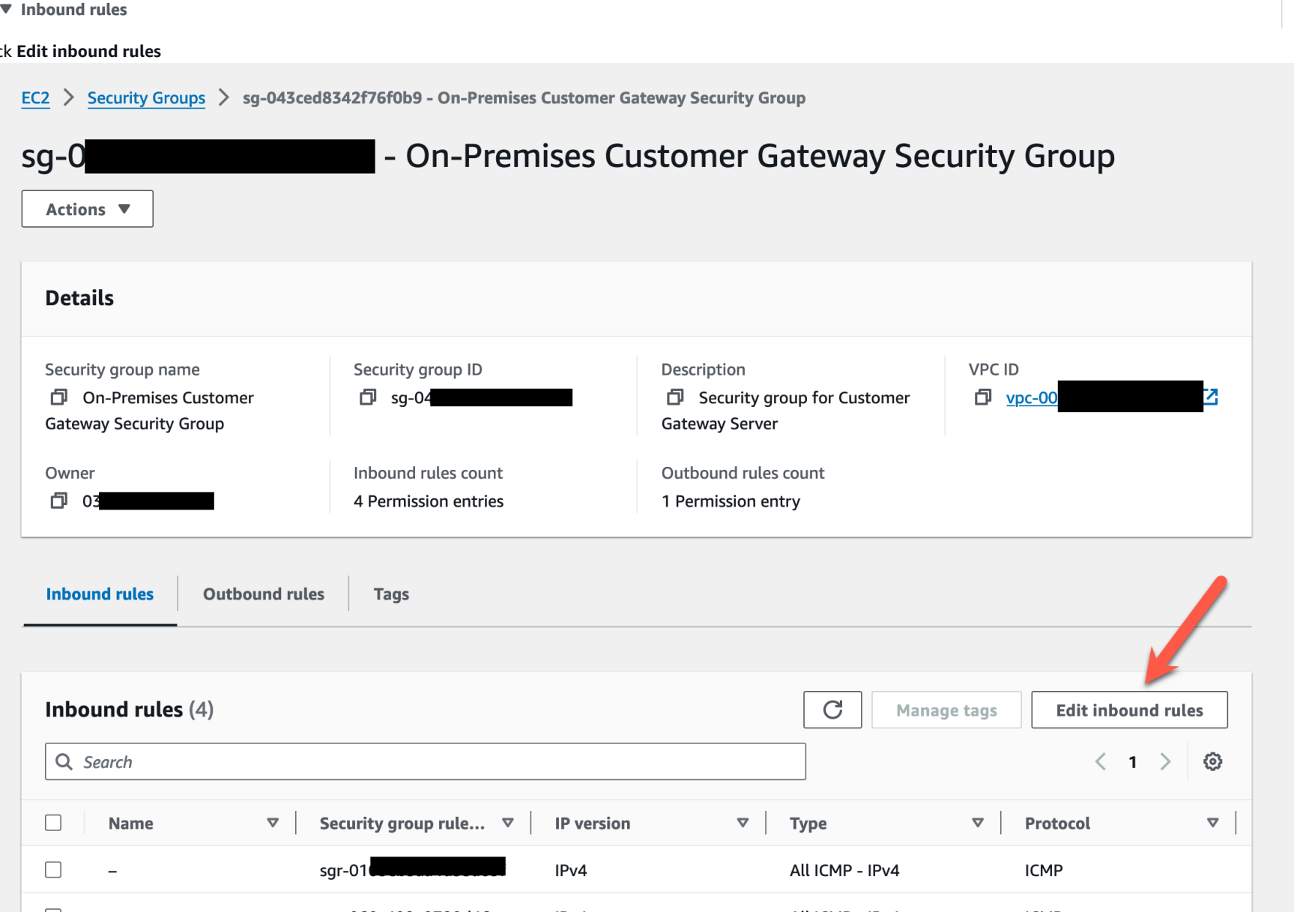
Owner ID: 0[redacted]

Launch time: Fri Mar 08 2024 11:24:03 GMT+0800 (Australian Western Standard Time)

Security groups: sg-043c[redacted] (On-Premises Customer Gateway Security Group)

▾ Inbound rules

3. Click **Edit inbound rules**



EC2 > Security Groups > sg-043ced8342f76f0b9 - On-Premises Customer Gateway Security Group

sg-0[redacted] - On-Premises Customer Gateway Security Group

Actions ▼

Details

Security group name: On-Premises Customer Gateway Security Group

Security group ID: sg-04[redacted]

Description: Security group for Customer Gateway Server

VPC ID: vpc-00[redacted]

Owner: 03[redacted]

Inbound rules count: 4 Permission entries

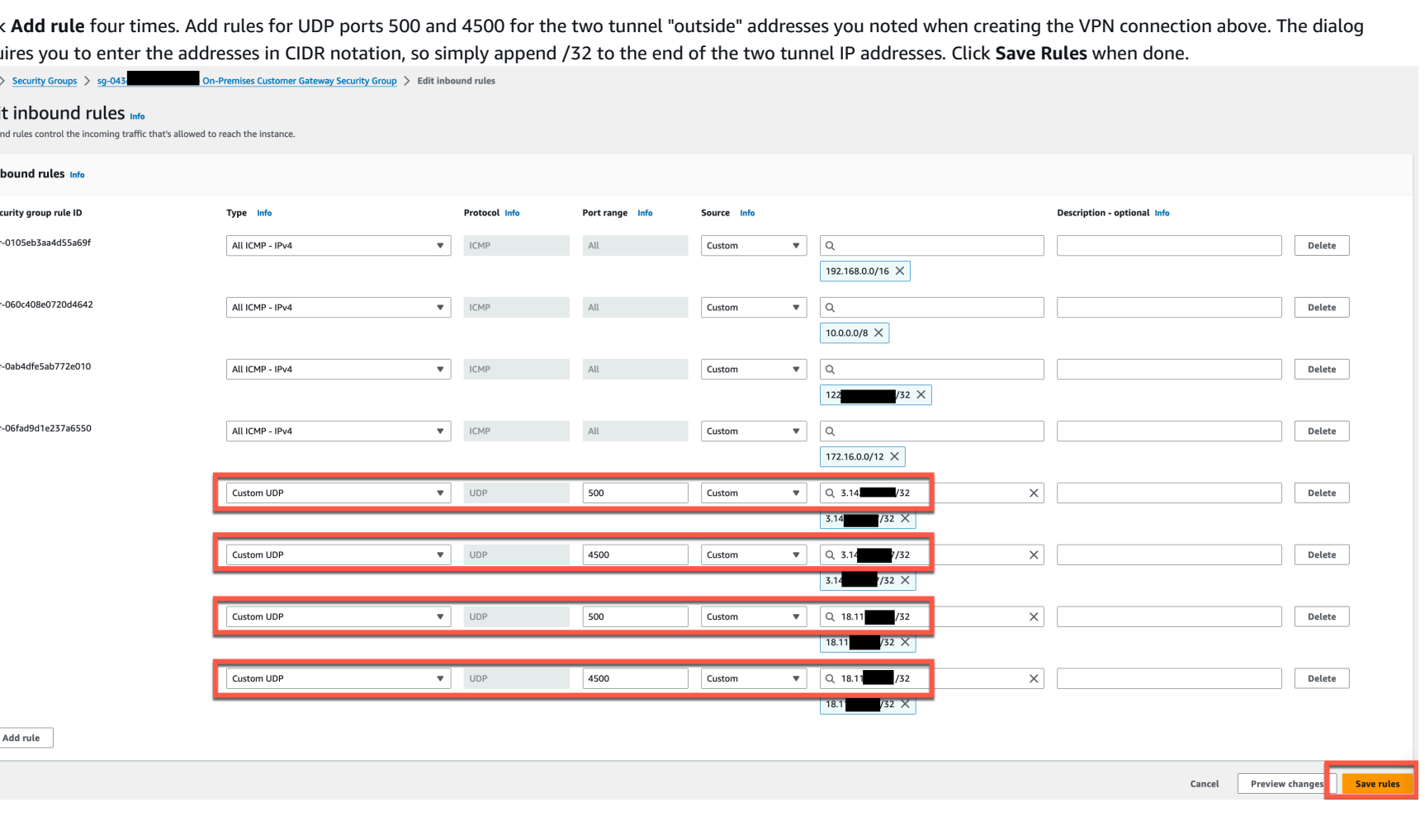
Outbound rules count: 1 Permission entry

Inbound rules | Outbound rules | Tags

Inbound rules (4) — Manage tags | Edit inbound rules

Name	Security group rule...	IP version	Type	Protocol
–	sgr-01[redacted]	IPv4	All ICMP - IPv4	ICMP

4. Click **Add rule** four times. Add rules for UDP ports 500 and 4500 for the two tunnel "outside" addresses you noted when creating the VPN connection above. The dialog requires you to enter the addresses in CIDR notation, so simply append /32 to the end of the two tunnel IP addresses. Click **Save Rules** when done.



EC2 > Security Groups > sg-043[redacted] On-Premises Customer Gateway Security Group > Edit inbound rules

Edit inbound rules Info

Inbound rules control the incoming traffic that's allowed to reach the instance.

Security group rule ID	Type	Protocol	Port range	Source		Description - optional	
sgr-0105eb3aa4d55a69f	All ICMP - IPv4	ICMP	All	Custom	192.168.0.0/16		Delete
sgr-060c408e0720d4642	All ICMP - IPv4	ICMP	All	Custom	10.0.0.0/8		Delete
sgr-0ab4dfe5ab772e010	All ICMP - IPv4	ICMP	All	Custom	12[redacted]/32		Delete
sgr-06fad9d1e257a6550	All ICMP - IPv4	ICMP	All	Custom	172.16.0.0/12		Delete
–	Custom UDP	UDP	500	Custom	3.14[redacted]/32		Delete
–	Custom UDP	UDP	4500	Custom	3.1[redacted]/32		Delete
–	Custom UDP	UDP	500	Custom	18.1[redacted]/32		Delete
–	Custom UDP	UDP	4500	Custom	18.1[redacted]/32		Delete

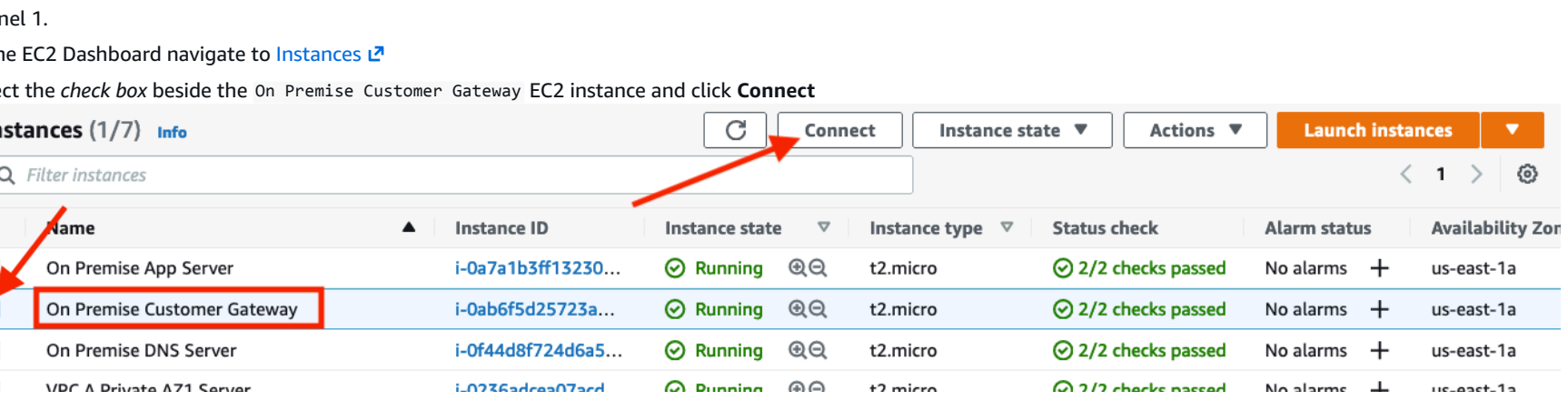
Add rule

Cancel | Preview changes | Save rules

Configure OpenSWAN and bring up the tunnel

Now that we have configured the simulated datacenter VPC and created the VPN connection to the Transit Gateway, we are ready to configure OpenSWAN on the Customer Gateway and bring up the tunnel. OpenSWAN has already been installed on the host during deploymennt and we will use the configuration file downloaded previously to configure the VPN. Since OpenSWAN does not provide built-in tunnel failover capability, we will only be configuring one of the tunnels.

1. Open the configuration file that you downloaded from the VPC console for the customer gateway in a text editor. We will be following this file to configure OpenSWAN for Tunnel 1.
2. In the EC2 Dashboard navigate to Instances
3. Select the *check box* beside the `On Premise Customer Gateway` EC2 instance and click **Connect**



Instances (1/7) Info

	Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zor
☐	On Premise App Server	i-0a7a1b3ff13230...	Running	t2.micro	2/2 checks passed	No alarms	us-east-1a
☑	On Premise Customer Gateway	i-0ab6f5d25723a...	Running	t2.micro	2/2 checks passed	No alarms	us-east-1a
☐	On Premise DNS Server	i-0f44d8f724d6a5...	Running	t2.micro	2/2 checks passed	No alarms	us-east-1a
☐	VPC A Private AZ1 Server	i-0236adcea07acd	Running	t2.micro	2/2 checks passed	No alarms	us-east-1a

4. Click **Connect** on the **Session manager** tab to open a shell.
5. Create an aws.conf file in `/etc/ipsec.d`:

```
sudo vim /etc/ipsec.d/aws.conf
```

Enter `i` to edit in vim.

▸ **Previous way of configuring OpenSWAN**

▾ **New way of configuring OpenSWAN**

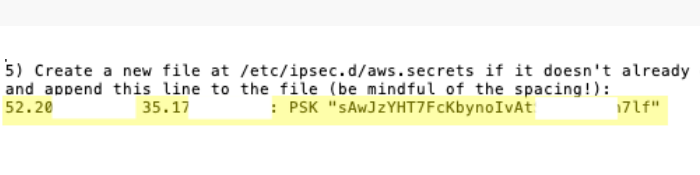
7. Copy and paste the following configuration. Modify `X.X.X.X` with your specific gateway IP address and `Y.Y.Y.Y` with your specific tunnel 1 IP addresses. You can get these IP addresses from the configuration file that you downloaded in Create Site-to-Site VPN Connection.

```
conn Tunnel1
    authby=secret
    auto=start
    left=%defaultroute
    leftid=X.X.X.X
    right=Y.Y.Y.Y
    type=tunnel
    ikelifetime=8h
    keylife=1h
    phase2alg=aes_gcm
    ike=aes256-sha2_256;dh14
    keyingtries=%forever
    keyexchange=ike
    leftsubnet=172.16.0.0/16
    rightsubnet=10.0.0.0/14
    dpddelay=10
    dpdtimeout=30
    dpdaction=restart_by_peer
```

8. Enter `esc` and `:wq!` to save the updates and exit vim.
9. Create an aws.secrets file in `/etc/ipsec.d`

```
sudo vim /etc/ipsec.d/aws.secrets
```

10. Copy and paste the pre-shared secret line from `Step 5` of the instructions into the secrets file.

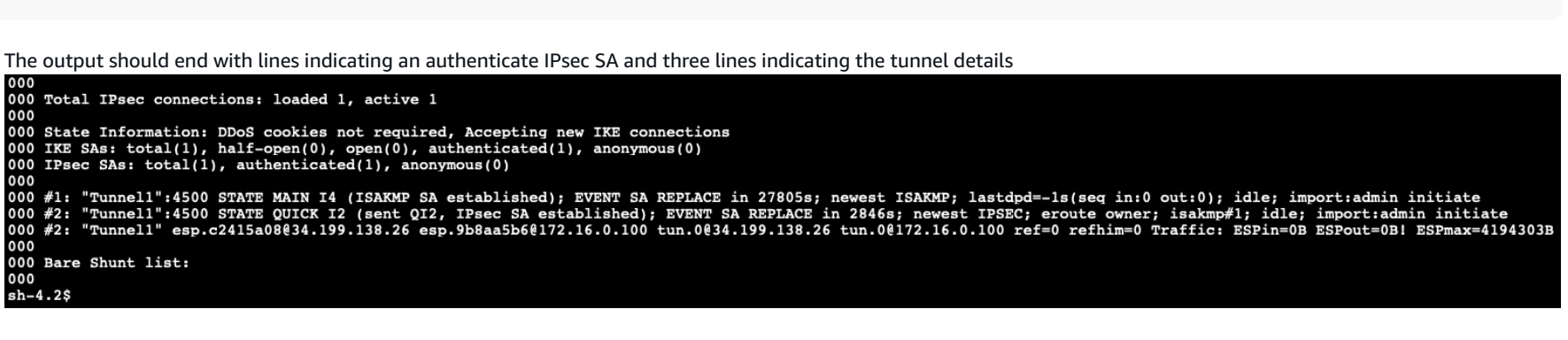


```
5) Create a new file at /etc/ipsec.d/aws.secrets if it doesn't already exist,
and append this line to the file (be mindful of the spacing!):
52.20          35.17          : PSK "sAwJzYHT7FcKbynoIvAt          i7lf"
```

11. Enter `esc` and `:wq!` to save the update to aws.secrets and exit vim.
12. Enable and start OpenSWAN:

```
sudo systemctl enable ipsec.service
sudo systemctl start ipsec.service
sudo ipsec status
```

13. The output should end with lines indicating an authenticate IPsec SA and three lines indicating the tunnel details



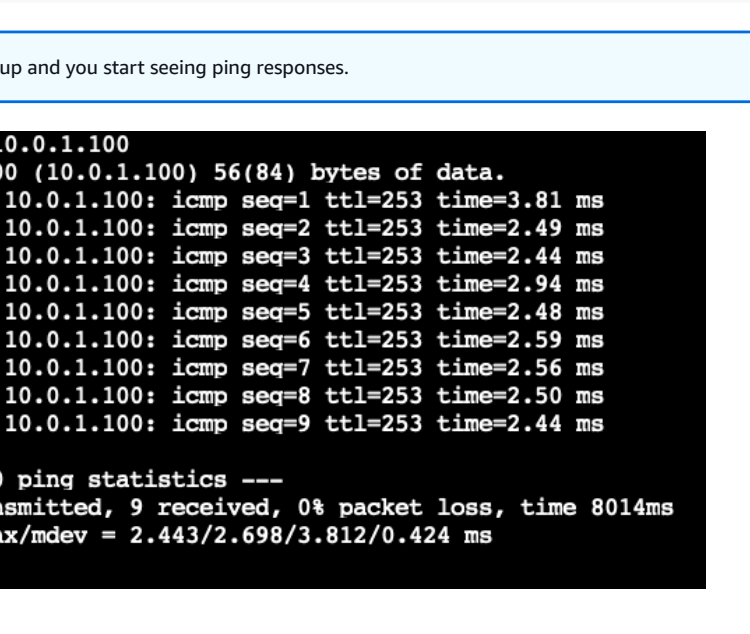
```
000
000 Total IPsec connections: loaded 1, active 1
000
000 State Information: DDoS cookies not required, Accepting new IKE connections
000 IKE SAs: total(1), half-open(0), open(0), authenticated(1), anonymous(0)
000 IPsec SAs: total(1), authenticated(1), anonymous(0)
000
000 #1: "Tunnel1":4500 STATE_MAIN_I4 (ISAKMP SA established); EVENT_SA_REPLACE in 27805s; newest ISAKMP; lastdpd=-1s(seq in:0 out:0); idle; import:admin initiate
000 #2: "Tunnel1":4500 STATE_QUICK_I2 (sent QI2, IPsec SA established); EVENT_SA_REPLACE in 2846s; newest IPSEC; eroute owner; isakmp#1; idle; import:admin initiate
000 #2: "Tunnel1" esp.c2415a08@34.199.138.26 esp.9b8aa5b6@172.16.0.100 tun.0@34.199.138.26 tun.0@172.16.0.100 ref=0 refhim=0 Traffic: ESPin=0B ESPout=0B! ESPmax=4194303B
000
000 Bare Shunt list:
000
sh-4.2$
```

> ⓘ AWS VPN connections are only brought up from the customer-side when using IKEv1, so we will need to send some traffic to one of our AWS VPCs from the simulated OnPrem environment.

14. From the Customer Gateway ping the private IP address of the EC2 instance in the private subnet in VPC A:

```
ping 10.0.1.100
```

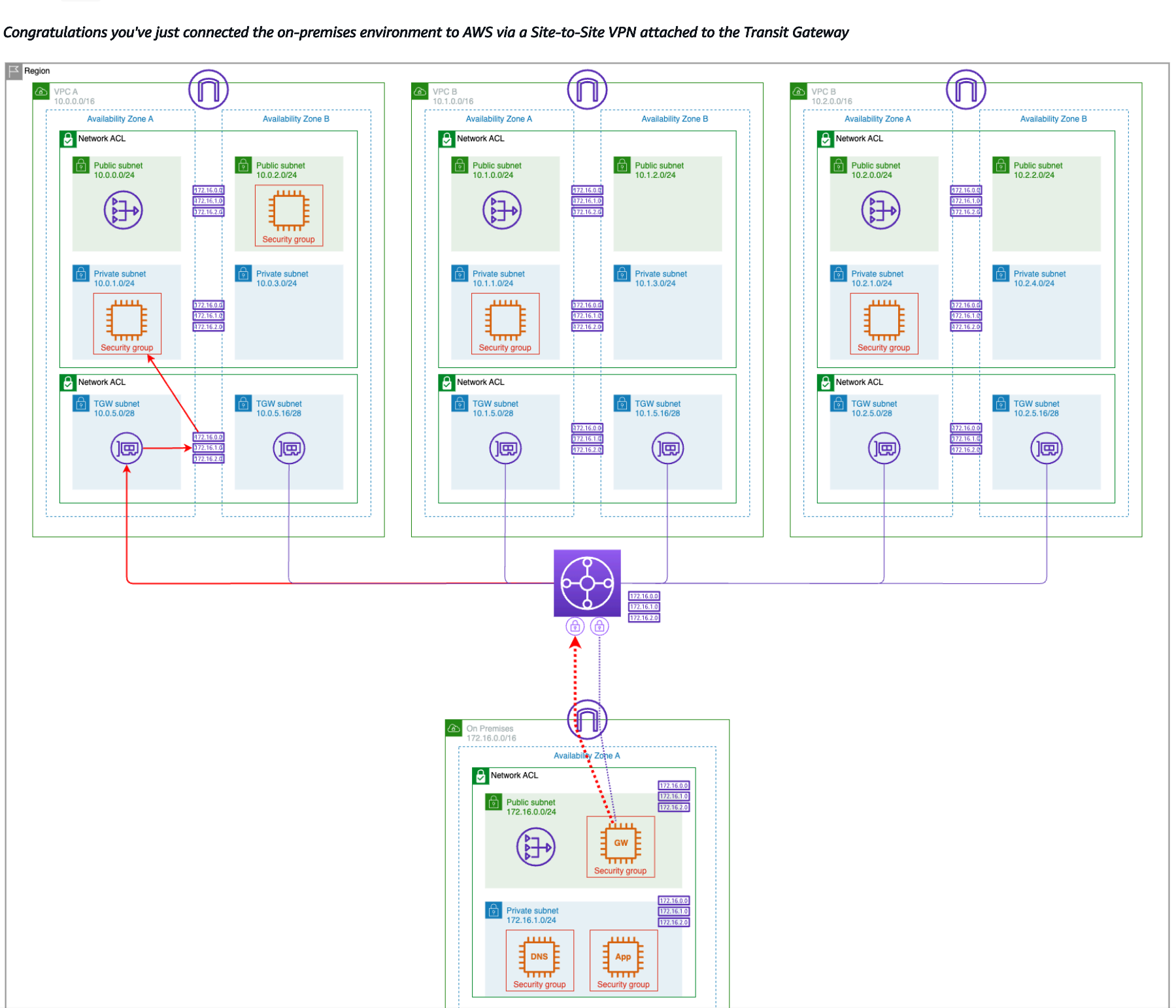
> ⓘ Note that it may take up to 30 seconds before the tunnel comes up and you start seeing ping responses.



```
sh-4.2$ ping 10.0.1.100
PING 10.0.1.100 (10.0.1.100) 56(84) bytes of data.
64 bytes from 10.0.1.100: icmp_seq=1 ttl=253 time=3.81 ms
64 bytes from 10.0.1.100: icmp_seq=2 ttl=253 time=2.49 ms
64 bytes from 10.0.1.100: icmp_seq=3 ttl=253 time=2.44 ms
64 bytes from 10.0.1.100: icmp_seq=4 ttl=253 time=2.94 ms
64 bytes from 10.0.1.100: icmp_seq=5 ttl=253 time=2.48 ms
64 bytes from 10.0.1.100: icmp_seq=6 ttl=253 time=2.59 ms
64 bytes from 10.0.1.100: icmp_seq=7 ttl=253 time=2.56 ms
64 bytes from 10.0.1.100: icmp_seq=8 ttl=253 time=2.50 ms
64 bytes from 10.0.1.100: icmp_seq=9 ttl=253 time=2.44 ms
^C
--- 10.0.1.100 ping statistics ---
9 packets transmitted, 9 received, 0% packet loss, time 8014ms
rtt min/avg/max/mdev = 2.443/2.698/3.812/0.424 ms
sh-4.2$
```

15. Enter `CTL C` to kill the ping command and terminate the Session Manager connection.

Congratulations you've just connected the on-premises environment to AWS via a Site-to-Site VPN attached to the Transit Gateway



Select your cookie preferences

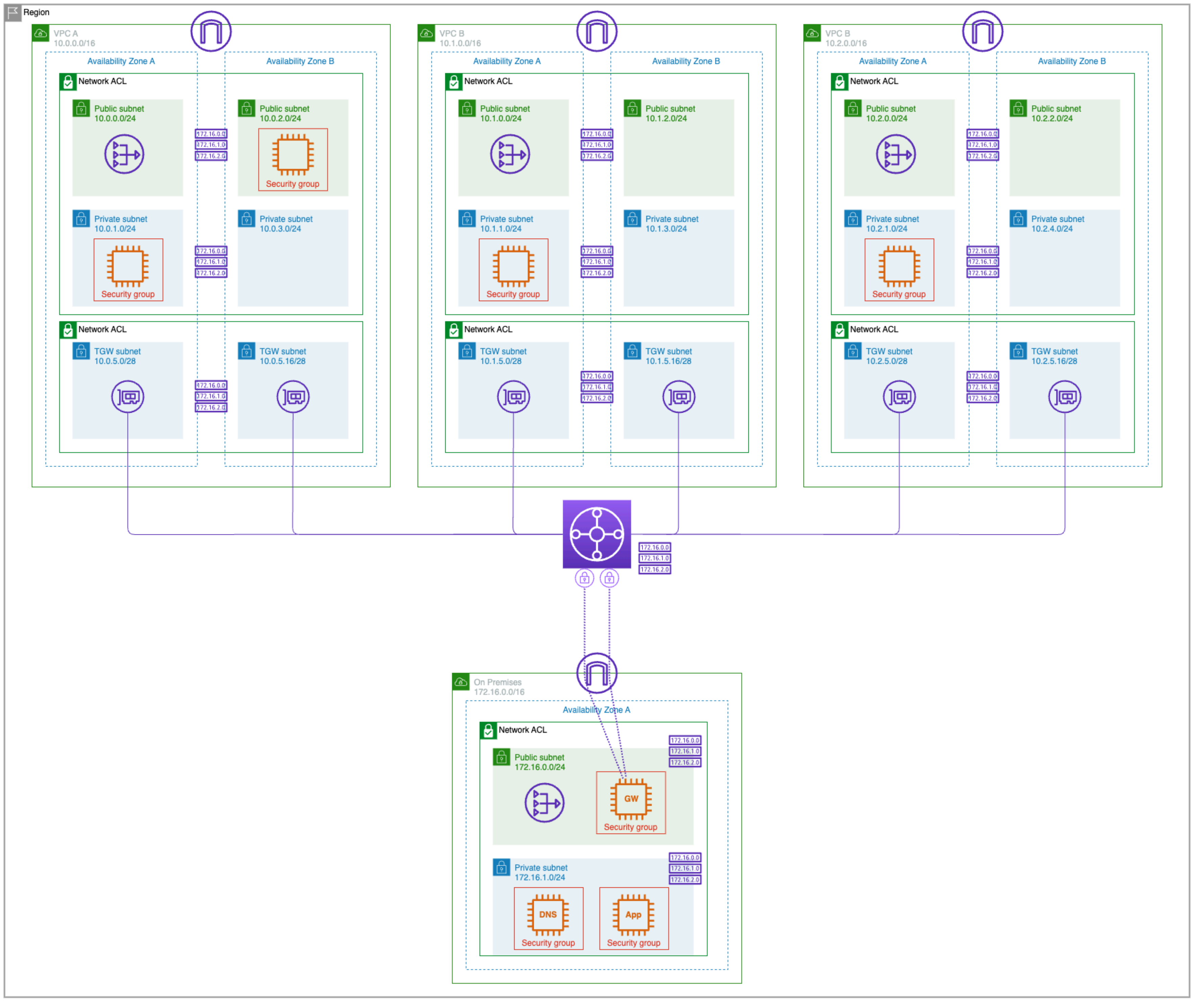
We use essential cookies and similar tools that are necessary to provide our site and services. We use performance cookies to collect anonymous statistics, so we can understand how customers use our site and make improvements. Essential cookies cannot be deactivated, but you can choose "Customize" or "Decline" to decline performance cookies.

If you agree, AWS and approved third parties will also use cookies to provide useful site features, remember your preferences, and display relevant content, including relevant advertising. To accept or decline all non-essential cookies, choose "Accept" or "Decline." To make more detailed choices, choose "Customize."

Accept | Decline | Customize

Explore Hybrid Environment

Now we are ready to explore the simulated on-premises environment.


Region
VPC A
10.0.0.0/16
Availability Zone A
Availability Zone B
Network ACL
Public subnet
10.0.0.0/24
Public subnet
10.0.2.0/24
Security group
Private subnet
10.0.1.0/24
Private subnet
10.0.3.0/24
Security group
Network ACL
TGW subnet
10.0.5.0/28
TGW subnet
10.0.5.16/28
172.16.0.0
172.16.1.0
172.16.2.0
VPC B
10.1.0.0/16
Public subnet
10.1.0.0/24
Public subnet
10.1.2.0/24
Private subnet
10.1.1.0/24
Private subnet
10.1.3.0/24
TGW subnet
10.1.5.0/28
TGW subnet
10.1.5.16/28
VPC B
10.2.0.0/16
Public subnet
10.2.0.0/24
Public subnet
10.2.2.0/24
Private subnet
10.2.1.0/24
Private subnet
10.2.4.0/24
TGW subnet
10.2.5.0/28
TGW subnet
10.2.5.16/28
On Premises
172.16.0.0/16
Availability Zone A
Network ACL
Public subnet
172.16.0.0/24
GW
Security group
Private subnet
172.16.1.0/24
DNS
Security group
App
Security group

Explore the simulated datacenter environment

We've now launched a simulated datacenter environment that consists of a customer gateway, a web application server, and a DNS server. Let's make sure that the components are working before we move on to connecting this to our AWS environment.

1. On the Customer Gateway instance check which DNS server the host is using by examining the `/etc/resolv.conf` file.

```
cat /etc/resolv.conf
```

2. Take note of the "nameserver" line, which should point to 172.16.1.200 which is the IP address of the on premise DNS Server created by CloudFormation template rather than the default VPC name server on 172.16.0.2.

```
sh-4.2$ cat /etc/resolv.conf
; generated by /usr/sbin/dhclient-script
search example.corp
options timeout:2 attempts:5
nameserver 172.16.1.200
```

3. Test the application server. In our simulated datacenter environment, we use the internal domain name *example.corp* and the application server has a hostname entry for *myapp.example.corp* created in the DNS server. We can test that the application server is up and running by using the curl command:

```
 curl http://myapp.example.corp
```

4. If your application server is working properly, you will see a response of "Hello, world."

```
sh-4.2$ curl http://myapp.example.corp
Hello, world.
```

5. Terminate the Session Manager connection.

Test the VPN connection from AWS to the simulated datacenter network.

Now that our VPN tunnel is up, we can test connectivity from the AWS environment, via the Transit Gateway, to our simulated on-premises environment. Recall that our on-premises environment contains an application server and a DNS server. The application server has a DNS A record in the on-premises DNS server for the **myapp.example.corp** hostname. In this section, we will verify connectivity between the AWS and simulated datacenter environments and test that we can connect to the application server.

1. In the EC2 Dashboard select Instances
2. Select the *check box* next to `VPC A Private AZ1 Server` and click **Connect** to start a Session Manager console
3. Test connectivity to the on premise application server's private IP address of 172.16.1.100

```
ping 172.16.1.100 -c 5
```

```
sh-4.2$ ping 172.16.1.100 -c 5
PING 172.16.1.100 (172.16.1.100) 56(84) bytes of data.
64 bytes from 172.16.1.100: icmp_seq=1 ttl=253 time=2.90 ms
64 bytes from 172.16.1.100: icmp_seq=2 ttl=253 time=2.28 ms
64 bytes from 172.16.1.100: icmp_seq=3 ttl=253 time=2.07 ms
64 bytes from 172.16.1.100: icmp_seq=4 ttl=253 time=2.17 ms
64 bytes from 172.16.1.100: icmp_seq=5 ttl=253 time=2.28 ms

--- 172.16.1.100 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4005ms
rtt min/avg/max/mdev = 2.077/2.345/2.909/0.293 ms
```

```
curl http://172.16.1.100
```

If your VPN connection is functioning correctly, this command will return "Hello, world." from the application server.

```
sh-4.2$ curl http://172.16.1.100
Hello, world.
```

5. Next let's verify that the AWS EC2 instances can reach the on-premises DNS server if we query it directly by using the IP address and the dig command to query for myapp.example.corp:

```
dig @172.16.1.200 myapp.example.corp
```

Your query should return an A record for the on-premises application server, similar to the following:

```
; <<>> DiG 9.11.4-P2-RedHat-9.11.4-26.P2.amzn2.5.2 <<>> @172.16.1.200 myapp.example.corp
; (1 server found)
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 2272
;; flags: qr aa rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 1, ADDITIONAL: 2

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 4096
;; QUESTION SECTION:
;myapp.example.corp.            IN      A

;; ANSWER SECTION:
myapp.example.corp.     60      IN      A       172.16.1.100

;; AUTHORITY SECTION:
example.corp.           86400   IN      NS      ns1.example.corp.

;; ADDITIONAL SECTION:
ns1.example.corp.       60      IN      A       172.16.1.200

;; Query time: 7 msec
;; SERVER: 172.16.1.200#53(172.16.1.200)
;; WHEN: Tue Sep 21 05:01:56 UTC 2021
;; MSG SIZE  rcvd: 97
```

6. Test the application server by using the hostname in the curl request:

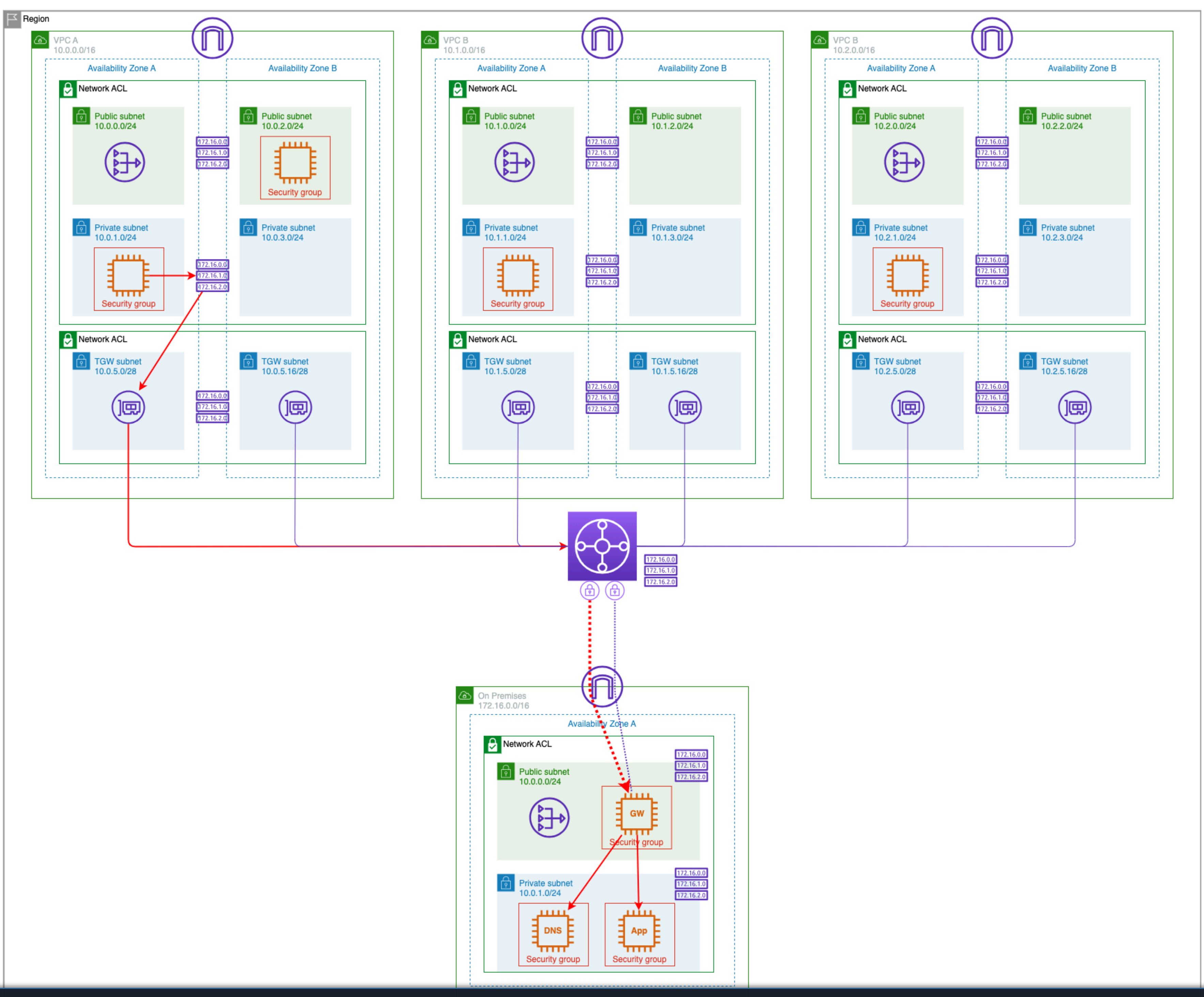
```
curl http://myapp.example.corp
```

⚠ **This should fail**. We will receive an error message "Could not resolve host" because DNS has not yet been configured to allow the AWS instances to resolve names hosted on the datacenter's DNS server.

```
sh-4.2$ curl http://myapp.example.corp
curl: (6) Could not resolve host: myapp.example.corp
```

7. Terminate the Session Manager connection.

Congratulations you've shown that there is connectivity in place from AWS to both the application server and DNS instance running in the on-premise environment. We will address hybrid-DNS in the next section that will allow DNS resolution of on-premise hosts from AWS.


Region
VPC A
10.0.0.0/16
Availability Zone A
Availability Zone B
Network ACL
Public subnet
10.0.0.0/24
Public subnet
10.0.2.0/24
Security group
Private subnet
10.0.1.0/24
Private subnet
10.0.3.0/24
Security group
Network ACL
TGW subnet
10.0.5.0/28
TGW subnet
10.0.5.16/28
172.16.0.0
172.16.1.0
172.16.2.0
VPC B
10.1.0.0/16
Public subnet
10.1.0.0/24
Public subnet
10.1.2.0/24
Private subnet
10.1.1.0/24
Private subnet
10.1.3.0/24
TGW subnet
10.1.5.0/28
TGW subnet
10.1.5.16/28
VPC B
10.2.0.0/16
Public subnet
10.2.0.0/24
Public subnet
10.2.2.0/24
Private subnet
10.2.1.0/24
Private subnet
10.2.3.0/24
TGW subnet
10.2.5.0/28
TGW subnet
10.2.5.16/28
On Premises
172.16.0.0/16
Availability Zone A
Network ACL
Public subnet
10.0.0.0/24
GW
Security group
Private subnet
10.0.1.0/24
DNS
Security group
App
Security group

Select your cookie preferences

We use essential cookies and similar tools that are necessary to provide our site and services. We use performance cookies to collect anonymous statistics, so we can understand how customers use our site and make improvements. Essential cookies cannot be deactivated, but you can choose "Customize" or "Decline" to decline performance cookies.

If you agree, AWS and approved third parties will also use cookies to provide useful site features, remember your preferences, and display relevant content, including relevant advertising. To accept or decline all non-essential cookies, choose "Accept" or "Decline." To make more detailed choices, choose "Customize."

Accept

Decline

Customize

AWS Networking Workshop

- Foundational Topics
  - Prerequisites
  - VPC Fundamentals
  - Multiple VPCs
  - Security Controls
  - Connecting to On-Premises
    - Establish VPN Connection
    - Explore Hybrid Environment
    - Hybrid DNS
    - Clean Up
  - Connecting to SDWAN
  - Network Monitoring
  - Network analytics using Amazon Quick Suite
- Advanced Topics
- Contributors & Revision History

Workshop catalog in AWS Builder Center

Hybrid DNS

Route 53 Resolver makes hybrid cloud easier for enterprise customers by enabling seamless DNS query resolution across your entire hybrid cloud. You can create DNS endpoints and conditional forwarding rules to allow resolution of DNS namespaces between your on-premises data center and AWS VPCs.

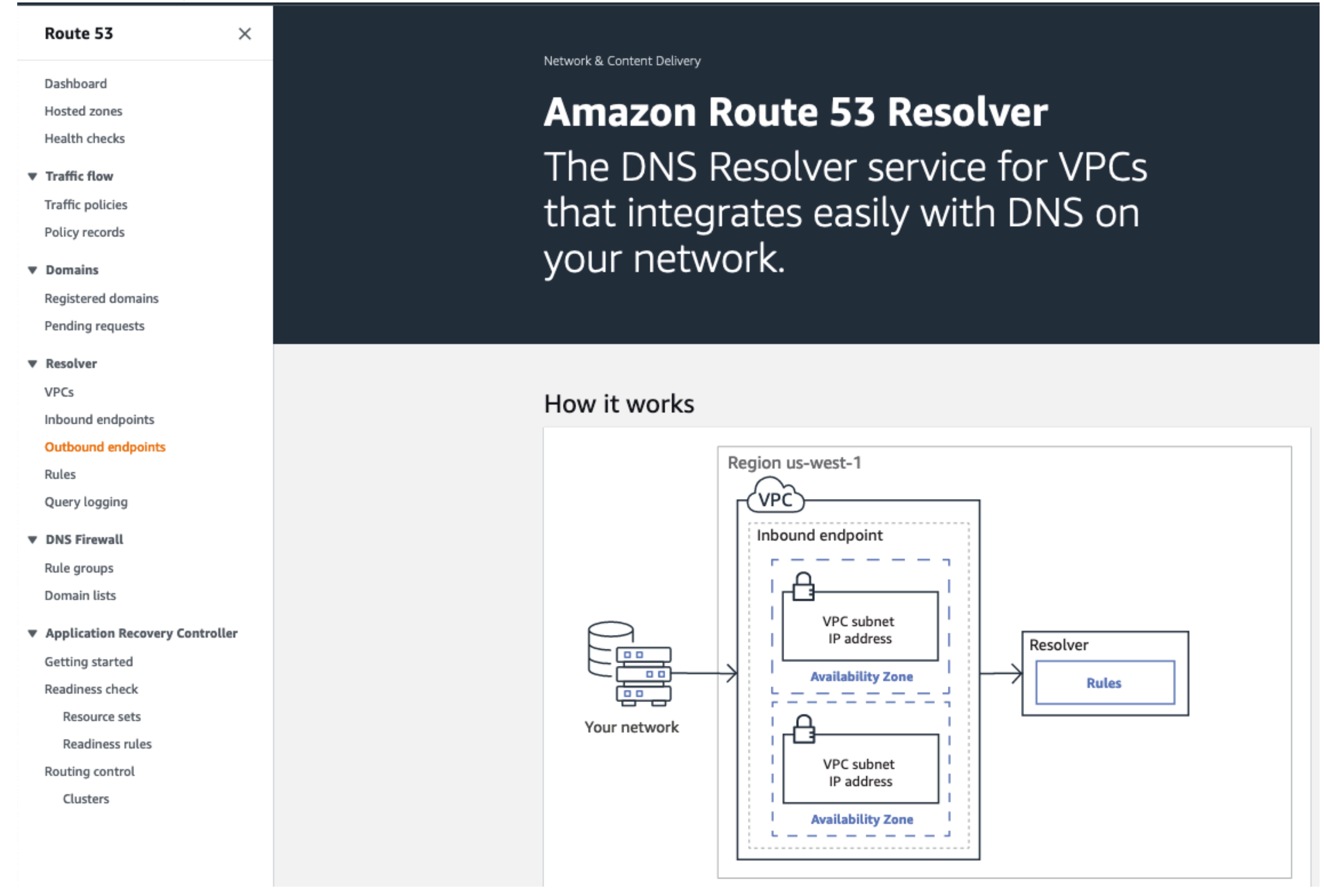
Recall that our simulated on-premises datacenter has a DNS server, providing authoritative name service for the example.corp domain where all internal application hosts are registered. In order to provide a complete hybrid-connectivity solution, we want to enable hosts in our AWS VPCs to resolve names of hosts in the datacenter environment. This can be achieved using Route 53 resolvers and conditional forwarding rules for the example.corp domain, while allowing the AWS instances to continue to take advantage of the highly available Amazon DNS service for all other name resolution inside the VPC and the internet.

For this exercise, we will focus on establishing DNS resolution from the AWS environment to the simulated datacenter, but it's important to note that the reverse is possible as well. Route 53 Resolvers supports inbound DNS queries that are conditionally forwarded from an on-premises DNS server. You can learn more about inbound DNS resolution in the AWS documentation.

Configure a Route 53 Resolver Outbound Endpoint

Route 53 Resolver uses endpoints to communicate with external DNS servers. An endpoint is an Elastic Network Interface (ENI) placed inside of a VPC which has connectivity to the existing DNS server. This may be a DNS server running on an EC2 instance, or a DNS server running on-premises accessible via Direct Connect or VPN. Since all three VPCs in our AWS environment have connectivity to the simulated datacenter via the Transit Gateway, we can use any of them for our endpoint. The endpoint will create interfaces in a minimum of two availability zones in your chosen VPC for high availability.

1. Navigate to Route 53,
2. Expand the hamburger on the top left and Under **Resolver** select **Outbound endpoints**

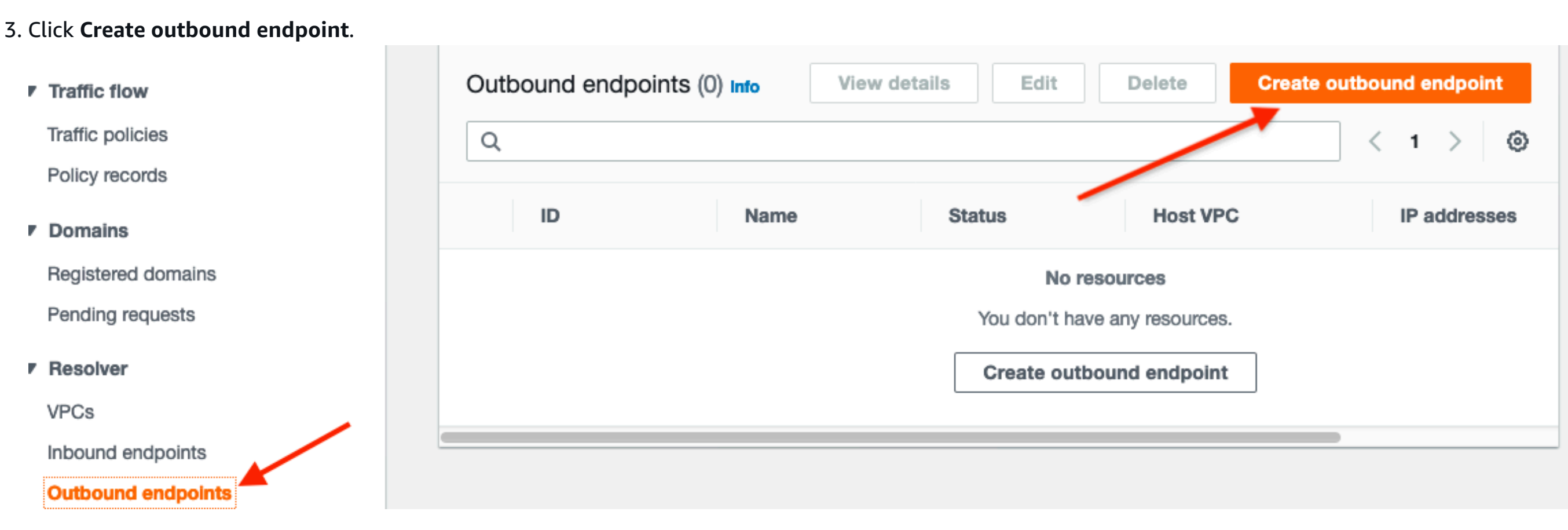

Route 53
×
Dashboard
Hosted zones
Health checks
Traffic flow
Traffic policies
Policy records
Domains
Registered domains
Pending requests
Resolver
VPCs
Inbound endpoints
Outbound endpoints
Rules
Query logging
DNS Firewall
Rule groups
Domain lists
Application Recovery Controller
Getting started
Readiness check
Resource sets
Readiness rules
Routing control
Clusters
Network & Content Delivery
How it works
Region us-west-1
VPC
Inbound endpoint
VPC subnet
IP address
Availability Zone
VPC subnet
IP address
Availability Zone
Your network
Resolver
Rules

Amazon Route 53 Resolver

The DNS Resolver service for VPCs that integrates easily with DNS on your network.

> ⓘ If you have not been to the Route 53 Resolver console before, it will take you to a splash page that you see in the screenshot above. Under **Resolver** on the left menu, select **Outbound endpoints** again to take you to the screen you see in the screenshot below. You want to see the button **Create outbound endpoint** not **Configure endpoints**

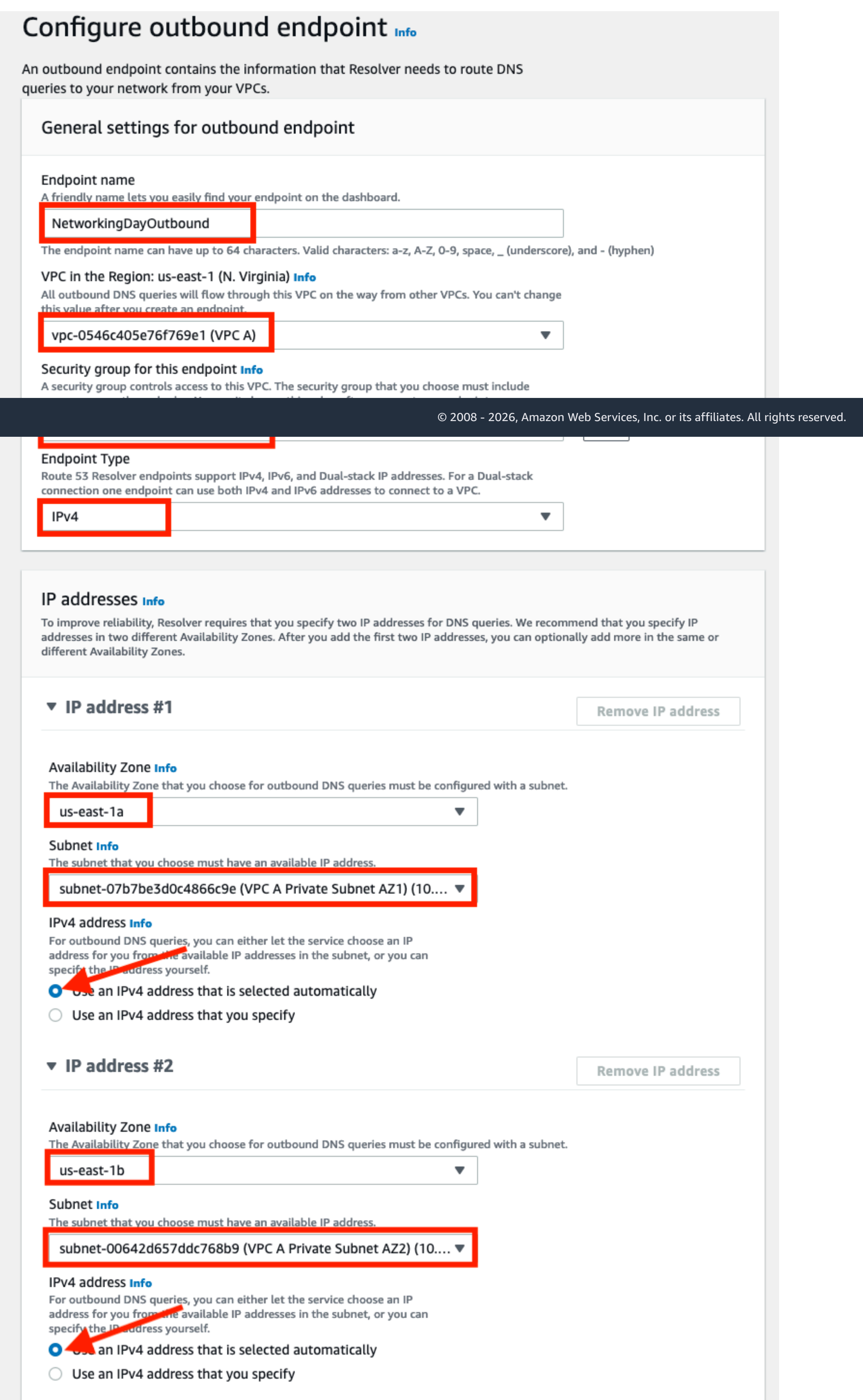
3. Click **Create outbound endpoint**.


Traffic flow
Traffic policies
Policy records
Domains
Registered domains
Pending requests
Resolver
VPCs
Inbound endpoints
Outbound endpoints
Outbound endpoints (0) Info
View details
Edit
Delete
Create outbound endpoint
1
ID
Name
Status
Host VPC
IP addresses
Create outbound endpoint

No resources

You don't have any resources.

4. Configure the settings for the outbound endpoint as follows:
   - **Endpoint name**: `NetworkingDayOutbound`
   - **VPC in the Region**: Select `VPC A`
   - **Security group for this endpoint**: Select the `default` Security Group for your VPC A, which already contains a rule permitting outbound connectivity.
   - **Endpoint type**: Select `IPv4`
   - **IP address #1**:
     - **Availability zone**: Choose `us-east-1a`
     - **Subnet**: `VPC A Private Subnet AZ1.`
     - **IP address**: `Use an IP address that is selected automatically`
   - **IP address #2**:
     - **Availability zone**: `us-east-1b`
     - **Subnet**: `VPC A Private Subnet AZ2.`
     - **IP address**: `Use an IP address that is selected automatically`



Configure outbound endpoint Info

An outbound endpoint contains the information that Resolver needs to route DNS queries to your network from your VPCs.

General settings for outbound endpoint

Endpoint name

A friendly name lets you easily find your endpoint on the dashboard.

NetworkingDayOutbound

The endpoint name can have up to 64 characters. Valid characters: a-z, A-Z, 0-9, space, _ (underscore), and - (hyphen)

VPC in the Region: us-east-1 (N. Virginia) Info

All outbound DNS queries will flow through this VPC on the way from other VPCs. You can't change this value after you create an endpoint.

vpc-0546c405e76f769e1 (VPC A) ▼

Security group for this endpoint Info

A security group controls access to this VPC. The security group that you choose must include

Endpoint Type

Route 53 Resolver endpoints support IPv4, IPv6, and Dual-stack IP addresses. For a Dual-stack connection one endpoint can use both IPv4 and IPv6 addresses to connect to a VPC.

IPv4 ▼

IP addresses Info

To improve reliability, Resolver requires that you specify two IP addresses for DNS queries. We recommend that you specify IP addresses in two different Availability Zones. After you add the first two IP addresses, you can optionally add more in the same or different Availability Zones.

▼ **IP address #1** Remove IP address

Availability Zone Info

The Availability Zone that you choose for outbound DNS queries must be configured with a subnet.

us-east-1a ▼

Subnet Info

The subnet that you choose must have an available IP address.

subnet-07b7be3d0c4866c9e (VPC A Private Subnet AZ1) (10.... ▼

IPv4 address Info

For outbound DNS queries, you can either let the service choose an IP address for you from the available IP addresses in the subnet, or you can specify the IP address yourself.

● Use an IPv4 address that is selected automatically

○ Use an IPv4 address that you specify

▼ **IP address #2** Remove IP address

Availability Zone Info

The Availability Zone that you choose for outbound DNS queries must be configured with a subnet.

us-east-1b ▼

Subnet Info

The subnet that you choose must have an available IP address.

subnet-00642d657ddc768b9 (VPC A Private Subnet AZ2) (10.... ▼

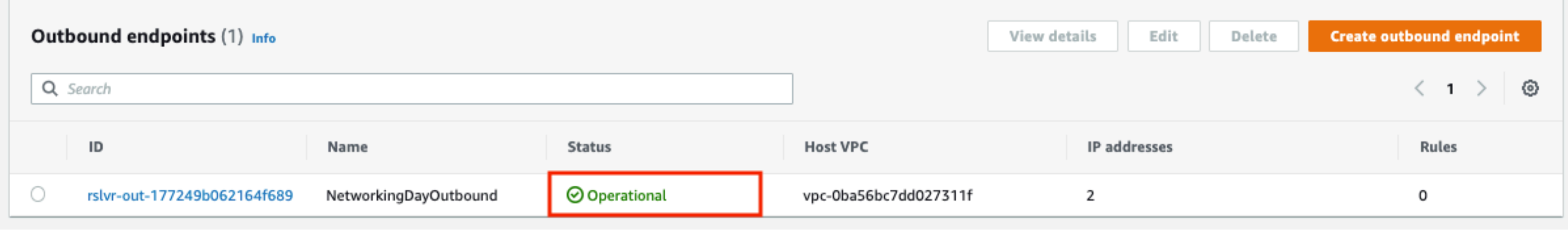
IPv4 address Info

For outbound DNS queries, you can either let the service choose an IP address for you from the available IP addresses in the subnet, or you can specify the IP address yourself.

● Use an IPv4 address that is selected automatically

○ Use an IPv4 address that you specify

5. Click **Submit** to create the endpoint.
6. Wait for the endpoint status to change to *Operational*. Then proceed to the next step.


View details
Edit
Delete
Create outbound endpoint
Search
1

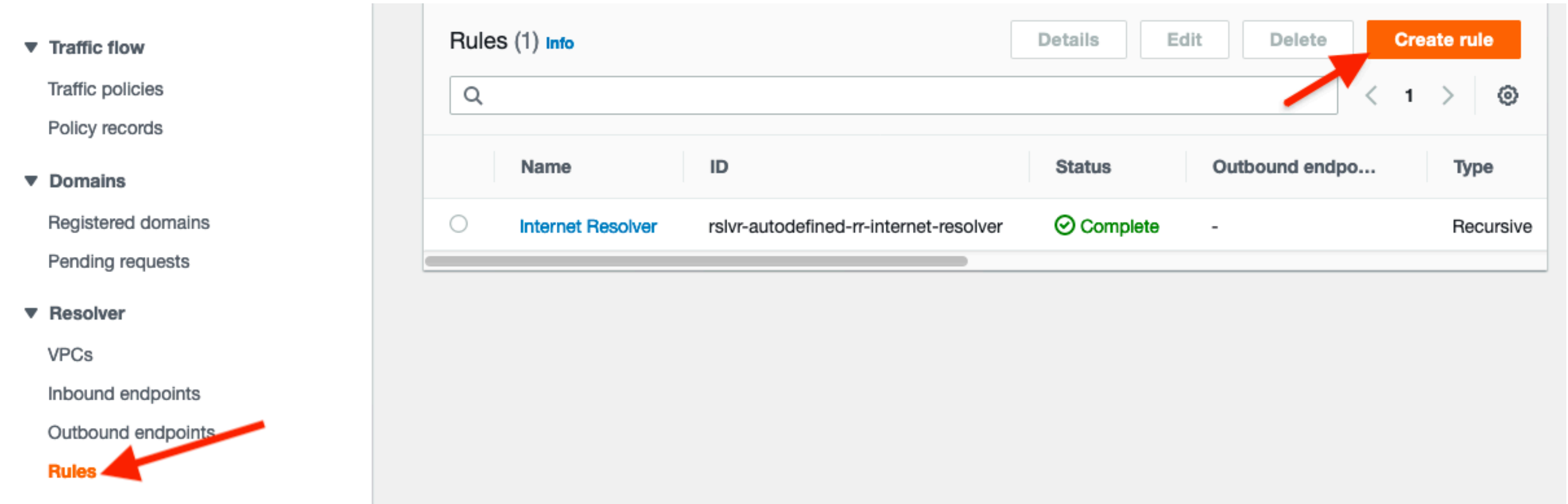
Outbound endpoints (1) Info

	ID	Name	Status	Host VPC	IP addresses	Rules
○	rslvr-out-177249b062164f689	NetworkingDayOutbound	Operational	vpc-0ba56bc7dd027311f	2	0

Create a Route 53 Resolver rule for example.corp.

Now that we have created an outbound endpoint this provides the Route 53 Resolver with a path to the on premise DNS server via VPC A and the Transit Gateway's VPN connection. Next we need to configure a *Route 53 Resolver Rule* to direct queries for **example.corp** to the on premise DNS server via that endpoint. We will associate the rule with VPC's A, B, and C and this will cause Route 53 Resolver to use this rule whenever the VPC DNS resolver processes queries for *example.corp* from instances in any of those three VPCs.

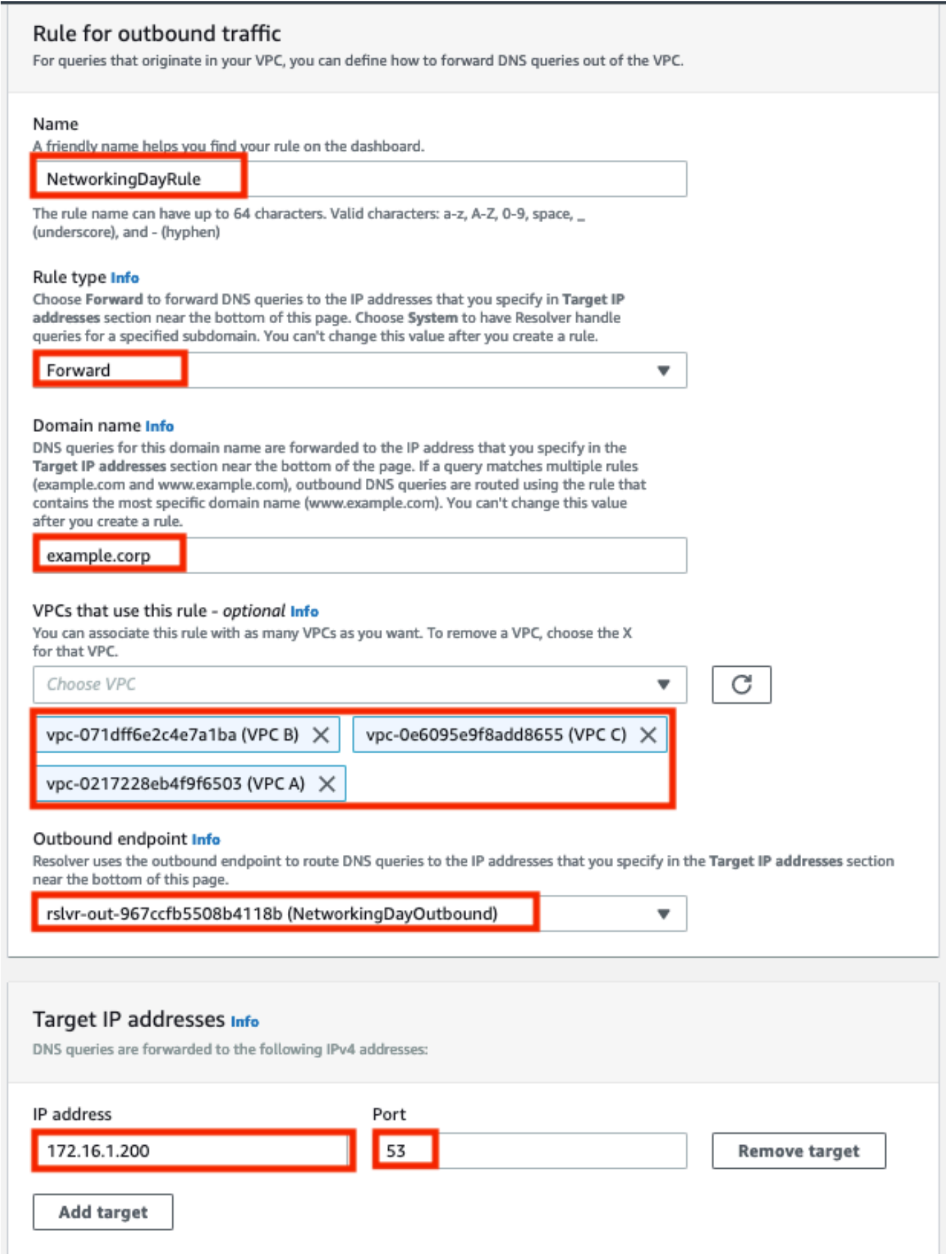
1. While still in the Route 53 console, navigate to the **Rules** tab under **Resolver**
2. Select **Create rule**


Traffic flow
Traffic policies
Policy records
Domains
Registered domains
Pending requests
Resolver
VPCs
Inbound endpoints
Outbound endpoints
Rules
Details
Edit
Delete
Create rule
1

Rules (1) Info

	Name	ID	Status	Outbound endpo...	Type
○	Internet Resolver	rslvr-autodefined-rr-internet-resolver	Complete	-	Recursive

3. Provide configuration for the rule:
   - Set **Name** as `NetworkingDayRule`
   - Leave **Rule type** as `Forward`
   - Specify the **Domain name** as `example.corp.`
   - Associate the rule with all three of your AWS VPCs: `VPC A`, `VPC B`, and `VPC C`.
   - For **Outbound endpoint** choose the endpoint you created in the previous step: `NetworkingDayOutbound`
   - Under **Target IP addresses** enter the on-premise DNS server IP address: `172.16.1.200`



Rule for outbound traffic

For queries that originate in your VPC, you can define how to forward DNS queries out of the VPC.

Name

A friendly name helps you find your rule on the dashboard.

NetworkingDayRule

The rule name can have up to 64 characters. Valid characters: a-z, A-Z, 0-9, space, _ (underscore), and - (hyphen)

Rule type Info

Choose **Forward** to forward DNS queries to the IP addresses that you specify in **Target IP addresses** section near the bottom of this page. Choose **System** to have Resolver handle queries for a specified subdomain. You can't change this value after you create a rule.

Forward ▼

Domain name Info

DNS queries for this domain name are forwarded to the IP address that you specify in the **Target IP addresses** section near the bottom of the page. If a query matches multiple rules (example.com and www.example.com), outbound DNS queries are routed using the rule that contains the most specific domain name (www.example.com). You can't change this value after you create a rule.

example.corp

VPCs that use this rule - *optional* Info

You can associate this rule with as many VPCs as you want. To remove a VPC, choose the X for that VPC.

Choose VPC ▼

vpc-071dff6e2c4e7a1ba (VPC B) ×   vpc-0e6095e9f8add8655 (VPC C) ×

vpc-0217228eb4f9f6503 (VPC A) ×

Outbound endpoint Info

Resolver uses the outbound endpoint to route DNS queries to the IP addresses that you specify in the **Target IP addresses** section near the bottom of this page.

rslvr-out-967ccfb5508b4118b (NetworkingDayOutbound) ▼

Target IP addresses Info

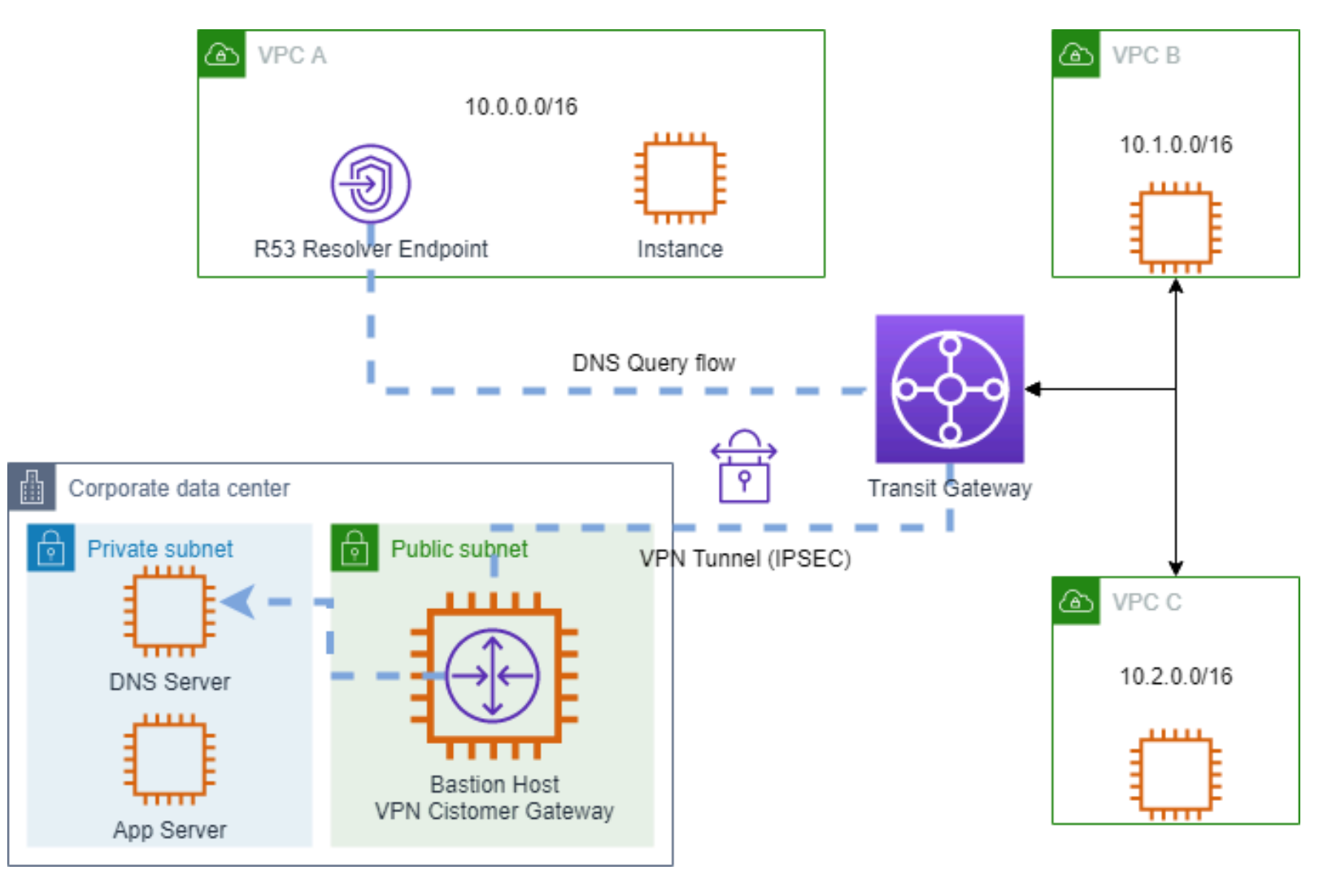
DNS queries are forwarded to the following IPv4 addresses:

IP address	Port	
172.16.1.200	53	Remove target

Add target

4. Click **Submit** to create the rule.

Testing the Route 53 Resolver rule


VPC A
10.0.0.0/16
R53 Resolver Endpoint
Instance
VPC B
10.1.0.0/16
DNS Query flow
Transit Gateway
VPN Tunnel (IPSEC)
Corporate data center
Private subnet
DNS Server
App Server
Public subnet
Bastion Host
VPN Cistomer Gateway
VPC C
10.2.0.0/16

At this point, we have configured Route 53 resolver to forward queries to the datacenter's DNS server for the example.corp domain from any of the AWS VPC's A, B, or C. We can test name resolution by attempting to connect to the on-premises app server from one of the EC2 instances in our AWS VPCs.

1. In the EC2 console navigate to Instances
2. Select the *check box* next to `VPC B Private AZ1 Server` and click the **Connect** button above.
3. On the **Session manager** tab click **Connect** to open a command prompt
4. Check setting of the DNS server on the instance by examining `resolv.conf`:

```
cat /etc/resolv.conf
```

Note that the instance is using the AWS provided DNS server (e.g. 10.1.0.2) and not your on-premises DNS server for name resolution:

```
nameserver 10.1.0.2
search ec2.internal
```

5. Check that the hostname for myapp.example.corp resolves to an IP address:

```
nslookup myapp.example.corp
```

Note that the instance is using the AWS provided DNS server (e.g. 10.1.0.2) and not your on-premises DNS server for name resolution:

```
nslookup myapp.example.corp
Server:         10.1.0.2
Address:        10.1.0.2#53

Non-authoritative answer:
Name:   myapp.example.corp
Address: 172.16.1.100
```

6. Use *curl* to query the on-premises application server by its hostname, myapp.example.corp

```
curl http://myapp.example.corp
```

If name resolution and your application server are functioning, you will receive back a "Hello, world" response.

```
curl http://myapp.example.corp
Hello, world.
```

7. Terminate the Session Manager connection.

Select your cookie preferences

We use essential cookies and similar tools that are necessary to provide our site and services. We use performance cookies to collect anonymous statistics, so we can understand how customers use our site and make improvements. Essential cookies cannot be deactivated, but you can choose "Customize" or "Decline" to decline performance cookies.

If you agree, AWS and approved third parties will also use cookies to provide useful site features, remember your preferences, and display relevant content, including relevant advertising. To accept or decline all non-essential cookies, choose "Accept" or "Decline." To make more detailed choices, choose "Customize."

Accept

Decline

Customize